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Thymic dendritic cell-derived IL-27p28 promotes the establishment of functional bias against IFN-γ production in newly generated CD4$^+$ T cells through STAT1-related epigenetic mechanisms

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eLife Assessment

This study presents a **useful** reassessment of the potential role of dendritic cell-derived IL-27 p28 cytokine in the functional maturation of CD4+CD8- thymocytes, and CD4+ recent thymic emigrants. The evidence supporting the claims of the authors is **solid** and serves to reaffirm what has been previously described, with the overall advance in understanding the mechanism(s) responsible for the intrathymic functional programming of CD4+ T cells being limited.

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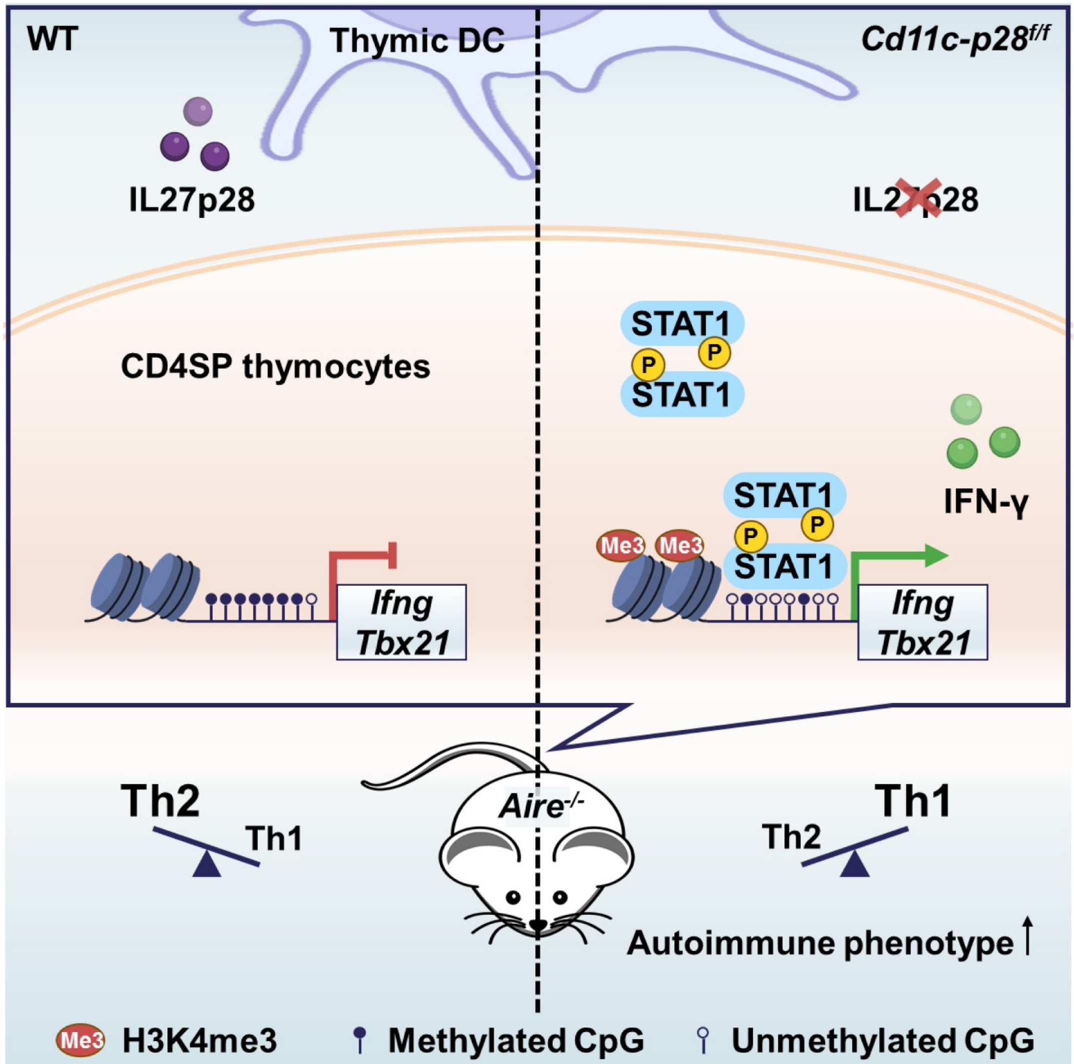
Abstract

The newly generated CD4 single-positive (SP) T lymphocytes are featured by enhanced IL-4 but repressed IFN-γ production. The mechanisms underlying this functional bias remain elusive. Previous studies have reported that CD4$^+$ T cells from mice harboring DC-specific deletion of IL-27p28 display an increased capacity of IFN-γ production upon TCR stimulation. Here we demonstrated that similarly altered functionality occurred in CD4SP thymocytes, recent thymic emigrants (RTEs), as well as naive T cells from either *Cd11c-p28*$^{f/f}$ mice or mice deficient in the α subunit of IL-27 receptor. Therefore, DC-derived IL-27p28-triggered, IL-27Rα-mediated signal is critically involved in the establishment of functional bias against IFN-γ production during their development in the thymus. Epigenetic analyses indicated reduced DNA methylation of the *Ifng* locus and increased trimethylation of H3K4 at both *Ifng* and *Tbx21* loci in CD4SP thymocytes from *Cd11c-p28*$^{f/f}$ mice. Transcriptome profiling demonstrated that *Il27p28* ablation resulted in the coordinated up-regulation of STAT1-activated genes. Concurrently, STAT1 was found to be constitutively activated. Moreover, we observed increased accumulation of STAT1 at the *Ifng* and *Tbx21* loci and a strong correlation

between STAT1 binding and H3K4me3 modification of these loci. Of note, *Il27p28* deficiency exacerbated the autoimmune phenotype of *Aire*$^{-/-}$ mice. Collectively, this study reveals a novel mechanism underlying the functional bias of newly generated CD4^{+} T cells and the potential relevance of such a bias in autoimmunity.

Graphical abstract



Introduction

The thymus plays a crucial role in the phenotypic development and functional maturation of T cells. After positive and negative selection, T cell progenitors become CD4 or CD8 single positive (CD4SP or CD8SP) thymocytes that are self-tolerant. These SP thymocytes then emigrate to the periphery, known as recent thymic emigrants (RTEs), and 2-3 weeks later, RTEs develop into mature naive T cells (Ashby & Hogquist, 2023). Upon activation, CD4SP thymocytes and RTEs exhibit a reduced proliferation capacity and decreased production of interleukin-2 (IL-2), interferon-γ (IFN-γ), and tumor necrosis factor α (TNF-α) compared to mature naive T cells. However, they show increased secretion of IL-4, which is known as T helper (Th) 2 bias in immature T cells (Hendricks & Fink, 2011). This Th2 bias in CD4SP thymocytes and RTEs is thought to assist newly emigrated self-reactive T cells to respond appropriately to self-antigens (Fink & Hendricks, 2011). Furthermore, the low level of IFN-γ production in RTEs renders them more prone to becoming regulatory T cells, which also helps to prevent the induction of autoimmune diseases (Bhaumik et al., 2013). The low level of DNA methylation at the *Il4* locus and reduced expression of *Dnmts* contribute to the Th2 bias in CD4SP thymocytes and RTEs (Berkley, Hendricks, Simmons, & Fink, 2013). However, the mechanism behind repressed IFN-γ expression in favor of the Th2 bias in these immature T cells is elusive.

The heterodimeric cytokine IL-27 is composed of IL-27p28 and Epstein-Barr virus-induced gene 3 (EBI3), which signals through the IL-27R consisting of the IL-27Rα (also named WSX-1, TCCR) and the gp130 subunits. Elevated levels of IL-27 are associated with the pathogenesis of autoimmune diseases such as ankylosing spondylitis (AS) (Lin et al., 2015), Behcet's disease (BD) (Shen, Xia, & Lu, 2013), and experimental arthritis (Cao, Doodes, Glant, & Finnegan, 2008). The IL-27-regulated differentiation and function of peripheral $CD4^+$ T cells subsets Th1, Th17 and Treg have been proposed as an explanation for the pathogenesis of autoimmune disease (Mei et al., 2021), however, conflicting reports have been demonstrated in different settings. For example, it is consistent that IL-27 inhibits Th17 differentiation and function *in vitro* and *in vivo* in *Il27p28*$^{-/-}$ and *Ebi3*$^{-/-}$ experimental autoimmune encephalomyelitis (EAE) mice models (Diveu et al., 2009; Kim et al., 2019; Liu et al., 2012). Although IL-27 inhibits Treg differentiation *in vitro* (Huber et al., 2008; Neufert et al., 2007), and *Ebi3*$^{-/-}$ EAE models show increased numbers and suppressive functions of Tregs (Liu et al., 2012), Tregs in *Il27ra*$^{-/-}$ and Treg-specific *Il27ra*$^{-/-}$ mice loss the suppressive function (Kim et al., 2019; Nguyen et al., 2019; Y. J. Park et al., 2019). Furthermore, IL-27 was initially found to promote Th1 differentiation of naive $CD4^+$ T cells under physiological conditions (Pflanz et al., 2002). However, naive $CD4^+$ T cells derived from *Ebi3*$^{-/-}$ and *Il27p28*$^{-/-}$ mice exhibit increased production of IFN-γ. In addition, the absence of *Il27ra*$^{-/-}$ promotes pathological Th1 accumulation following *Trypanosoma cruzi* infection (Hamano et al., 2003). A subsequent study demonstrates that both the absence and overexpression of IL-27p28 lead to increased susceptibility to *T. gondii* infection, highlighting the role of IL-27p28 as an independent negative regulator of IFN-γ production in T cells (J. Park et al., 2019; Villarino et al., 2003). Notably, deficiency in gp130 does not alter IFN-γ production in $CD4^+$ T cells in a virus-infection model (Harker, Wong, Dolgoter, & Zuniga, 2015). These results indicate the intricate functionality of each subunit of IL-27 and IL-27R in peripheral $CD4^+$ T cells, which complicates the interpretation of the autoimmune phenotype in deficient mice.

The structural analysis of IL-27 has demonstrated that IL-27p28, serving as a central subunit, possesses the ability to bind with EBI3, as well as the receptor subunits IL-27Rα (Caveney, Glassman, Jude, Tsutsumi, & Garcia, 2022). Previous studies have reported IL-27p28 as an independent antagonist of gp130-mediated signaling (Stumhofer et al., 2010). In a ConA-induced liver damage model, $CD4^+$ T cells from mice with dendritic cell (DC)-specific deletion of p28 (*Cd11c-p28*$^{f/f}$ mice) exhibit a propensity for rapid and robust production of IFN-γ upon activation. This observation is attributed to the absence of IL-27p28 in the thymic environment of these mice (Zhang et al., 2013). These findings raise the intriguing possibility that thymic DC-derived IL-27p28 may contribute to shaping the Th1/Th2 balance of newly generated CD4SP thymocytes.

In this paper, we have demonstrated that DC-derived IL-27p28 plays a crucial role in maintaining the Th2 bias of CD4SP thymocytes. Our findings suggest that IL-27p28 regulates the DNA methylation status and histone modifications in the transcription regulatory regions of *Ifng* and *Tbx21* genes. These epigenetic modifications were further modulated by the low level of basal and phosphorylated signal transducer and activator of transcription 1 (STAT1) protein. Furthermore, we observed that the disruption of Th2 bias in CD4SP thymocytes due to IL-27p28 deficiency exacerbated autoimmune diseases in *Aire*$^{-/-}$ mice. Our results provide insights into the molecular mechanisms underlying the regulation of Th2 bias by DC-derived IL-27p28 and its potential role in autoimmune diseases.

Results

1. DC-specific deletion of IL-27p28 endows the newly generated CD4⁺ T cells with inherently enhanced capacity of IFN-γ production

Although IL-27 is widely reported to promote peripheral Th1 differentiation *in vitro*, $CD4^+$ T cells from *Cd11c-p28*$^{f/f}$ mice exhibit enhanced IFN-γ production, a capacity seemingly acquired in their development in the thymus (Zhang et al., 2013). To investigate the possibility that thymic DC-derived IL-27 shapes the functionality of newly generated $CD4^+$ T cells, CD4SP thymocytes, $CD4^+$ RTEs, and $CD4^+$ naive T cells were purified from *Cd11c-p28*$^{f/f}$ (cKO) mice and their wild-type (WT) littermates harboring a Rag2-GFP transgene, which allows easy tracking of newly generated thymocytes and RTEs (Ashby & Hogquist, 2023). The purified cells were then compared for cytokine expression in response to anti-CD3 and anti-CD28 stimulation under non-polarizing conditions. As expected, $CD4^+$ naive T cells from cKO mice exhibited a significantly higher level of *Ifng* mRNA expression compared to WT cells (**Figure 1A**). Intracellular staining (**Figure 1B**) and ELISA assay of the culture supernatant (**Figure 1C**) confirmed enhanced IFN-γ production at the protein level. Notably, this difference was already detectable at the SP thymocyte stage, indicating that it is developmentally regulated. Another prominent feature of CD4SP thymocytes is the hyperproduction of Th2 cytokines (Fink & Hendricks, 2011; Makar et al., 2003). Despite the profound impact of IL-27p28 deficiency on IFN-γ production, we observed no significant difference in IL-4 expression between cKO and WT T cells (**Figure 1A** and **C**). Moreover, there was no significant difference in the expression of IL-2 and TNF-α between cKO and WT T cells (**Figure 1A** and Figure 1-figure supplement 1). These results suggest that DC-specific deletion of IL-27p28 specifically enhances IFN-γ production in newly generated T cells without affecting Th2 cytokines.

Next, we investigated the expression of T-box transcription factor (T-bet) and GATA3, which are the master regulators of Th1 and Th2 differentiation, respectively (Fang, Healy, & Zhu, 2022). The differential impact of IL-27p28 deficiency on IFN-γ and IL-4 production was accompanied by an increase in Tbx21 expression but no significant alteration in Gata3 expression (**Figure 1D**). Western blotting and flow cytometry confirmed the upregulation of T-bet protein expression (**Figure 1E, F**). Furthermore, we extended our analyses onto $CD4^+$ T cells at steady state or under polarizing conditions. Higher basal levels of *Ifng* and *Tbx21* transcripts were detected in CD4SP thymocytes and naive T cells from cKO mice than the counterparts from WT mice even without stimulation (**Figure 1G**), indicating the constitutive activation of these loci. When cultured under polarizing conditions, these cells were effectively induced to differentiate into Th1, Th2, Th17 and Treg lineages. In contrast to the previous report of different differentiation potentials of RTEs versus naive T cells (Hendricks & Fink, 2011), CD4SP thymocytes, RTEs and naive T cells showed similar capacity to differentiate into effector cells. Moreover, cKO and WT cells were equally positioned for the differentiation (Figure 1-figure supplement 2A-F), suggesting that IL-27p28 deficiency does not affect effector cell differentiation under optimal conditions. Moreover, the frequency of IFN-γ-, and granzyme B-producing $CD8^+$ T cells under Th0 conditions was comparable between cKO and WT T cells (Figure 1-figure supplement 2G). Taken together, these data support that IL-27p28 is actively involved in the functional maturation of developing CD4SP thymocytes, biasing them away from the Th1 lineage.

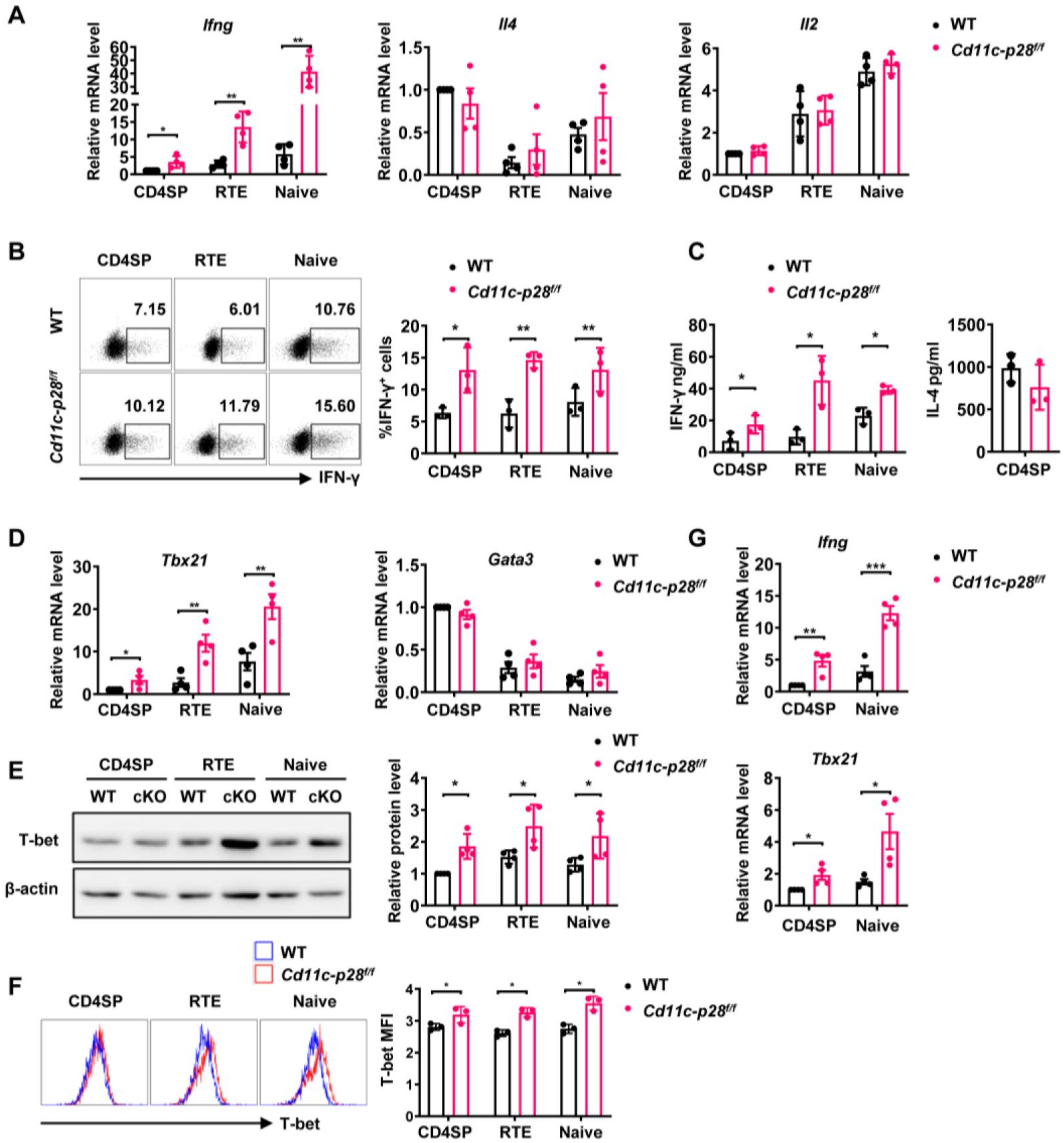


Figure 1.

Elevated IFN-γ production and T-bet expression in *Cd11c-p28*$^{f/f}$ mice initiated from CD4SP thymocytes stage.

(A) CD4SP ($GFP^+CD4^+CD8^-CD44^{lo}$) thymocytes, $CD4^+$ RTEs ($GFP^+CD4^+CD8^-CD25^-CD44^{lo}$) and $CD4^+$ naive ($GFP^-CD4^+CD8^-CD25^-CD44^{lo}$) T cells were sorted from 6-8-week-old *Cd11c-p28*$^{f/f}$ mice and WT littermates, stimulated with plate coated anti-CD3 (2 μg/mL) and soluble anti-CD28 (1 μg/mL) for 12 hours. mRNA levels of *Ifng, Il4,* and *Il2* were determined by qPCR. Data: mean ±SD (n=4, duplicates).
(B) Sorted cells were cultured under Th0 conditions for 3 days. The frequency of IFN-γ-producing $CD4^+$ T cells were measured by intracellular staining. Representative dot plots (left) and statistical data (right, mean ±SD, n=3).
(C) Supernatants from 3-day cultures were analyzed for IFN-γ and IL-4 by ELISA. Data: mean ±SD (n=3).
(D) mRNA levels of *Tbx21* and *Gata3* in sorted cells were determined by qPCR. Data: mean ±SD (n=4, duplicates).
(E-F) T-bet protein levels were assessed by Western blot (E) and flow cytometry (F) after 3-day culture. Data: mean ±SD (n=3).
(G) Freshly sorted cells were lysed in Trizol, and *Ifng* and *Tbx21* mRNA levels were determined by qPCR. Data: mean ±SD (n=4, duplicates). Statistical differences: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ (Student's *t*-test).

2. *Il27ra*-deficient $CD4^+$ T cells are similarly predisposed for potent IFN-γ production

Apart from acting as a subunit of IL-27, IL-27p28 have been reported to possess activities independent of IL-27, by acting on its own or coupling with other proteins, such as IL-Y (p28/p40, binding to IL-27Rα/IL-12Rβ1 to activate STAT3) (Flores et al., 2015) and CLY (p28/CLF, binding to gp130/IL-6Rα to activate STAT1 and STAT3) (Tormo et al., 2013). To ascertain whether the phenotype in the *Cd11c-p28*$^{f/f}$ mice results from defective IL-27Rα signaling, we investigated the functional characters of T lineage cells in mice deficient in *Il27ra*. Reminiscent to what was observed in *Il27p28*-deficient mice, increased expression of *Ifng* and *Tbx21* were detected in *Il27ra*$^{-/-}$ CD4SP thymocytes and naive $CD4^+$ T cells, whereas *Il4*, *Il2,* and *Gata3* transcripts were equally present in WT and *Il27ra*$^{-/-}$ cells (**Figure 2A**). Intracellular staining confirmed the enhanced IFN-γ production by *Il27ra*-deficient cells stimulated under non-polarizing conditions (**Figure 2B**). IFN-γ production by *Il27ra*$^{-/-}$ and WT cells, however, was comparable under Th1 polarizing conditions (Figure 1-figure supplement 2H). The similar phenotype shared by *Il27p28* and *Il27ra* knockout mice indicates that the altered T cell functionality in the absence of IL-27p28 is indeed a reflection of disrupted IL-27Rα signaling.

3. IL-27p28 deficiency induces permissive epigenetic changes at *Ifng* and *Tbx21* loci

Developmentally regulated transcriptional activation and repression of genes or cell-type specific expression patterns are largely achieved by modifying the chromatin template at a gene locus (Ivashkiv, 2018). Murine neonatal $CD4^+$ T cells have been documented to be poised to rapid Th2 cytokine secretion due to the pre-existing hypomethylation at a key Th2 cytokine regulatory region (Rose, Lichtenheld, Foote, & Adkins, 2007). In other reports, Th2 polarization in adult SP thymocytes and RTEs is attributable to active recruitment of DNA methyltransferases and increased H3K4 methylation at *Il4* locus (Berkley et al., 2013; Makar et al., 2003). In addition, DNA methylation and histone modification are found to be important for the control of *Ifng* and *Tbx21* expression (Fang et al., 2022; Friedman, Lee, Lee, & Oh, 2023; Jones & Chen, 2006). Therefore, we next evaluated the contribution of epigenetic modifications to the enhanced IFN-γ production in the absence of IL-27p28. Firstly, we performed bisulfite genomic sequencing on DNA of CD4 SP thymocytes from *Cd11c-cre p28*$^{f/f}$ mice and WT littermates. We focused on the nine CpG sites (**Figure 3A**) most proximal to the *Ifng* transcription start site, among which the −53 CpG has previously been suggested to play a key role in IFN-γ repression in Th2 effector cells (Jones & Chen, 2006). DNA methylation was significantly reduced at three CpG sites (−53, −34 and +16 site) of the *Ifng* locus in CD4SP thymocytes from*Cd11c-cre p28*$^{f/f}$ mice (**Figure 3B**). On the contrary, the five CpG sites in the *Il4* promoter region (**Figure 3C, D**), whose demethylation is associated with high level IL-4 production (Lee, Agarwal, & Rao, 2002), showed no difference between cKO and WT CD4SP thymocytes.

As much as histone modifications are concerned, histone H3 lysine 27 trimethylation (H3K27me3) and H3 lysine 9 trimethylation (H3K9me3) are typically associated with repressed *Ifng* and *Tbx21* expression in Th cells, whereas histone H3 lysine 4 trimethylation (H3K4me3) enhances gene expression (Fang et al., 2022; Friedman et al., 2023). To evaluate the histone trimethylation modifications at *Ifng*, *Tbx21*, *Il4* and *Gata3* loci in CD4SP thymocytes from WT and *Cd11c-p28*$^{f/f}$ mice, chromatin immunoprecipitation was performed using antibodies against H3K4me3, H3K9me3 and H3K27me3. H3K4me3 was found to be accumulated at *Ifng* and *Tbx21* loci in cKO CD4SP thymocytes, while its level in *Gata3* and *Il4* loci was comparable to that of wild type cells (**Figure 3E**). On the other hand, neither the occupancy of H3K9me3 and H3K27me3 at *Ifng*, *Tbx21*, *Il4* and *Gata3* loci was altered (**Figure 3F, G**), nor were the total protein levels of H3K4me3, H3K9me3 H3K27me3 or the mRNA level of enzymes catalyzing the formation of these

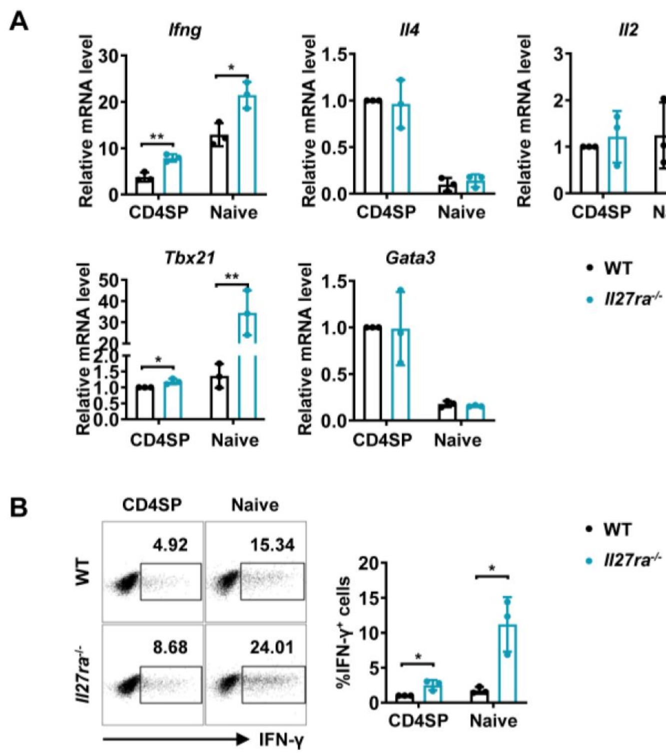


Figure 2.

Enhanced IFN-γ production and T-bet expression in *Il27ra*$^{-/-}$ mice initiated at the CD4SP thymocytes stage.

(A) CD4SP thymocytes and naive CD4^{+} T cells were isolated from *Il27ra*$^{-/-}$ and WT mice, stimulated with plate-coated anti-CD3 (2 µg/mL) and soluble anti-CD28 (1 µg/mL) for 12 hours. mRNA levels of *Ifng, Il4*, *Il2, Tbx21,* and *Gata3* were determined by qPCR. Data: mean ±SD (n=3, duplicates).

(B) CD4SP thymocytes and CD4^{+} naive T cells were cultured under Th0 conditions for 3 days. The frequency of IFN-γ-producing CD4^{+} T cells were analyzed by intracellular staining. Representative dot plots (left) and quantification (right, mean ± SD, n=3). Significance: * *P*< 0.05, ** *P*< 0.01 (Student's *t*-test).

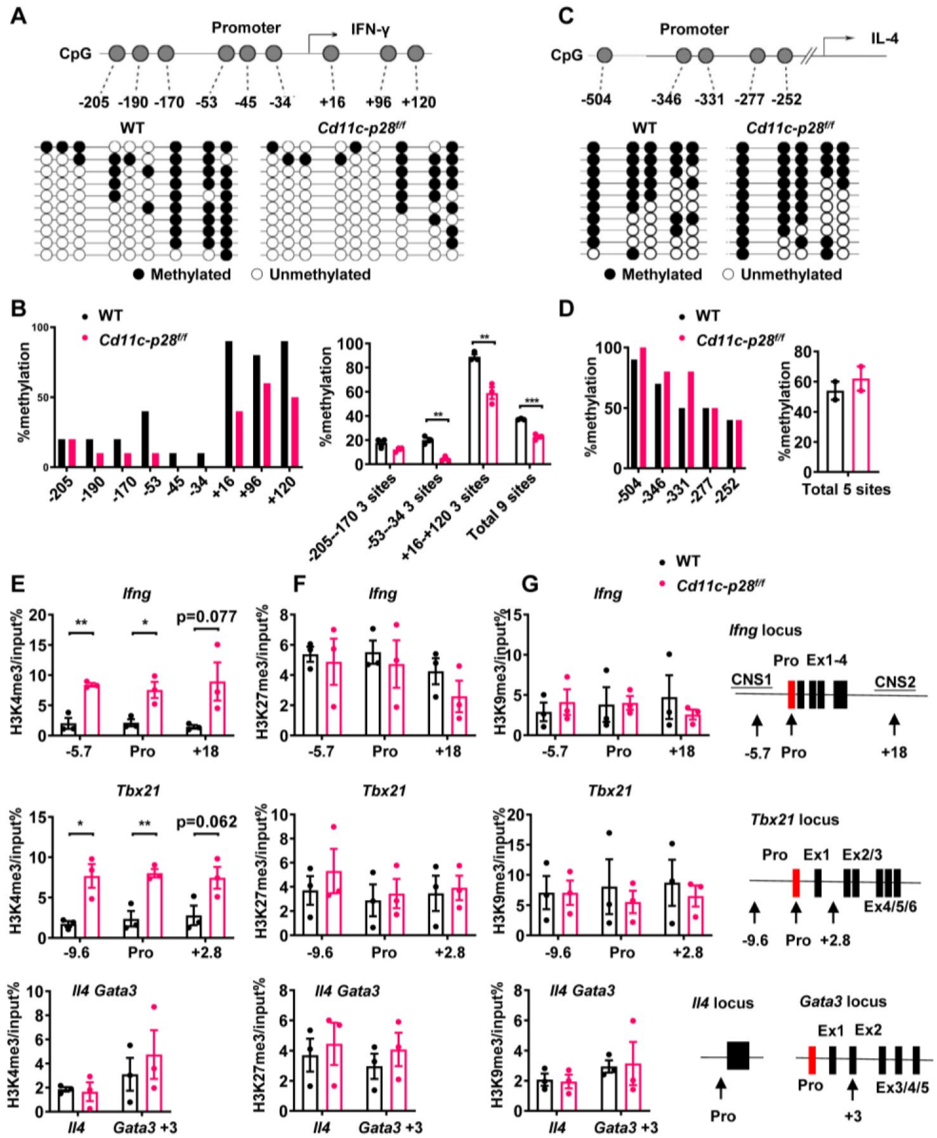


Figure 3.

Distinct DNA and H3K4 methylation patterns at *Ifng* and *Tbx21* promoter regions in CD4SP thymocytes from *IL27p28*-deficient mice.

(A) DNA methylation analysis of nine CpG sites in the *Ifng* promoter using sodium bisulfite-treated genomic DNA from GFP^{+}CD4^{+}CD8^{-}CD44^{lo} CD4SP thymocytes. Each row represents a sequenced allele (n=10 clones from one of the three independent experiments). Filled (●) and open (○) circles denote methylated and unmethylated cytosine, respectively.
(B) Left: Percent methylation at individual CpG sites from one representative experiment. Right: Average methylation of three adjacent site groups (group1: −205, − 190, −170; group2: −53, −45, −34; group3: +16, +96, +120) and all CpG sites (mean ± SD, n=3).
(C) DNA methylation analysis of five CpG sites upstream of the *Il4* transcription start site using sodium bisulfite-treated genomic DNA from GFP^{+}CD4^{+}CD8^{-}CD44^{lo} CD4SP thymocytes. Each row represents a sequenced allele (n=10 clones from the two independent experiments). Filled (●) and open (○) circles denote methylated and unmethylated cytosine, respectively.
(D) Left: Graphs show the percentage of methylation at each individual site (left panel) or all CpG sites (right panel).
(E-G) Histone trimethylation analysis in freshly isolated CD4SP thymocytes from *IL27p28*-deficient and WT mice. ChIP-qPCR was performed using antibodies against H3K4me3 (E), H3K27me3 (F), and H3K9me3 (G). qPCR primers targeted promoter and trans-regulatory regions of *Ifng*, *Tbx21*, *Il4,* and *Gata3*. Data: mean ±SEM (n=3, duplicates). Significance: * P<0.05; ** P<0.01 (Student's *t*-test). Abbreviation: pro., promoter.

modifications (Figure 3-figure supplement 1A-C). These results suggest a link between up-regulated *Ifng* and *Tbx21* expression and altered epigenetic modifications in the absence of IL-27p28.

4. Coordinated up-regulation of STAT1-activated genes in CD4SP thymocytes from *Cd11c-p28*$^{f/f}$ mice

To better understand the molecular mechanisms underlying the altered functionality, the transcriptional profile of CD4SP thymocytes from *Cd11c-p28*$^{f/f}$ mice was explored by RNA-Seq and compared with that of the WT counterparts. The two sets of differentially expressed genes (DEGs) were subjected to analyses for gene ontology (GO) biological process terms and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways. The up-regulated DEGs showed significant enrichment in genes related to host defense to viral infection, including interferon response and Jak-STAT signaling (**Figure 4A**). No meaningful hit, on the other hand, was revealed for the down-regulated gene set (data not shown).

Given that STAT1 is a key mediator downstream of IL-27 signaling (Philips et al., 2022), we were particularly interested in the altered expression of STAT1-regulated genes in the absence of IL-27p28. A previous study by Hirahara et al. (Hirahara et al., 2015) has compared the transcriptomes of wild-type and STAT1-deficient $CD4^+$ T cells stimulated by IL-27. Using their data (GSE65621), we compiled a list of genes potentially under regulation of STAT1. The DEGs were then examined for overlaps with this gene set. Surprisingly, over 50% of up-regulated DEGs (69 out of 137) fell into the category of STAT1-activated genes, whereas none of them belonged to STAT1-suppressed genes (**Figure 4B**). To validate the RNA-Seq data, qPCR was carried out for a group of overlapping genes, including *Gm12250, Oasl2, Oas3, Parp14, Ifit3, Usp18, Igtp, Irf1, Ifi44, Oas2, Rsad2, Il12r*. As shown in **Figure 4C**, all these genes showed differential expression between WT and cKO CD4SP thymocytes. To further illustrate modest but coordinate changes in the expression of STAT1-regulated genes, gene set enrichment analysis (GSEA) was performed without imposing an arbitrary fold-change cutoff. Indeed, significant enrichment of STAT1-activated genes was detected in the transcriptome of CD4SP thymocytes from the knockout mice (NES=1.67, NOM p-val=10^{-16}, **Figure 4D**). Therefore, IL-27p28 deficiency resulted in an enhanced STAT1 activity.

To reveal functionally important connections of the DEGs, protein-protein interaction network was generated using Network Analyst (Xia, Gill, & Hancock, 2015). After removal of isolated and loosely connected nodes, the rest of the proteins encoded by the DEGs were imported into Network Analyst for network construction. As shown in **Figure 4E**, STAT1 was centrally positioned in the network with the highest degree of interactions. Taken together, these results indicate that IL-27p28 deficiency is strongly associated with an altered transcriptional program featured by enhanced expression of STAT1-activated genes.

5. Constitutive activation of STAT1 in CD4SP thymocytes in the absence of IL-27p28

The distinct transcriptional signature prompted us to examine the activation status of STAT1 in freshly isolated CD4SP thymocytes from *Cd11c-p28*$^{f/f}$ mice and the WT littermates. Total thymocytes were stained for CD4 and CD8, followed by intracellular staining with antibodies against phosphorylated STATs. Significant levels of STAT1 Y701 phosphorylation were observed in cKO but not in WT cells. On the other hand, both cKO and WT cells were stained negative for phosphorylated STAT3 and STAT4 (**Figure 5A**). Phosphorylated and total STAT proteins were also analyzed by Western blotting in purified CD4SP thymocytes, RTEs and naive $CD4^+$ T cells. While the total protein of STAT1 was comparable between cKO and WT cells in all populations examined, elevated levels of phosphorylated STAT1 Y701 were constantly detected in cKO cells. Moreover, the phosphorylation of STAT1 Y701 appeared to be developmentally regulated, increasing progressively from CD4SP thymocytes to naive T cells (**Figure 5B**). Another predominant

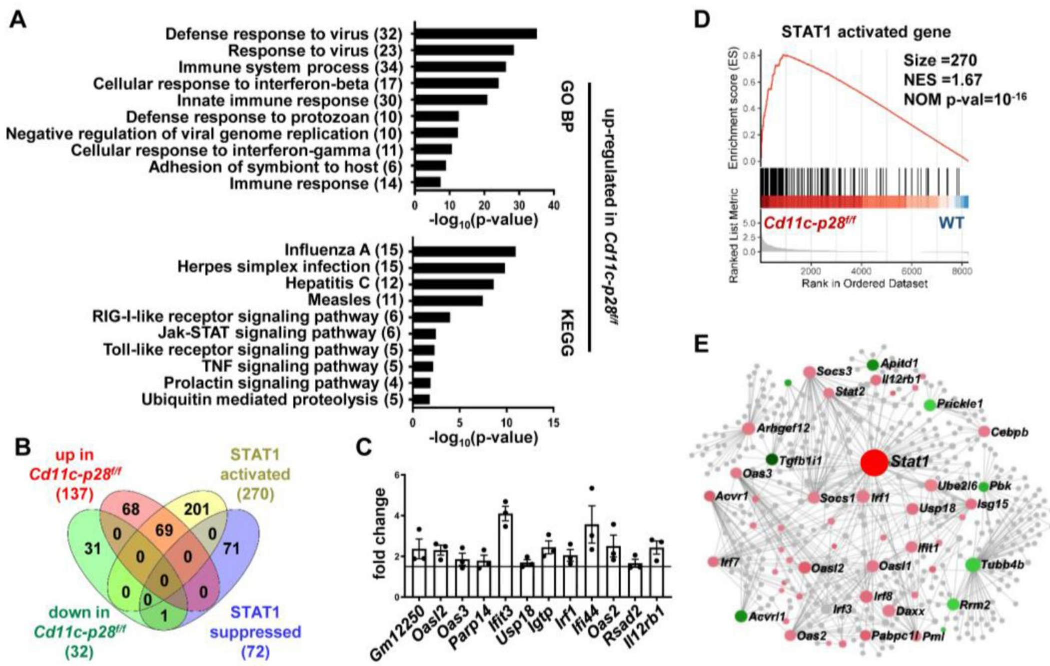


Figure 4.

Increased expression of STAT1-activated genes in CD4SP thymocytes from *Cd11c-p28*$^{f/f}$ mice.

RNA-seq was performed to analyze the transcriptome of CD4SP thymocytes from *Cd11c-p28*$^{f/f}$and WT mice.

(A) Top 10 enriched Gene Ontology (GO) biological process and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways for up-regulated differentially expressed genes (DEGs) in *Cd11c-p28*$^{f/f}$mice.

(B) Overlap between DEGs in *Cd11c-p28*$^{f/f}$ mice and STAT1-activated and suppressed genes. Numbers indicate overlapping genes in each category.

(C) Validation of RNA-Seq results by qPCR for representative up-regulated genes. Data: fold change in *Cd11c-p28*$^{f/f}$versus WT mice (mean ± SEM, n=3, duplicates).

(D) Gene Set Enrichment Analysis (GSEA) showing coordinated upregulation of STAT1-activated genes in *Cd11c-p28*$^{f/f}$CD4SP thymocytes.

(E) Protein-protein interaction network of DEGs. Nodes represent proteins; edges indicate interactions. Larger nodes denote higher interaction degrees. Red: up-regulated; green: down-regulated; gray: non-DEGs connected to the network (added by Network Analyst).

phosphorylation site in STAT1 is serine 727, which is crucial for maximal STAT1 transcription activity (Barnholt, Kota, Aung, & Rutledge, 2009). Basal levels of S727 phosphorylation were also observed in CD4SP thymocytes, but with no difference in WT and cKO cells (data not shown). As much as STAT3 and STAT4 were concerned, neither total nor phosphorylated STAT3 and STAT4 were found to be altered with IL-27p28 ablation (**Figure 5B**), indicating that the impact is specific for STAT1.

STAT1 plays an important role in the development of Th1 response by regulating *Tbx21* and *Ifng* expression (Fang et al., 2022). To explore the correlation of the increased STAT1 activation with the enhanced T-bet expression and IFN-γ production, we assessed STAT1 binding on the *Tbx21* and *Ifng* loci by ChIP assay. Possibly due to the relatively low abundance of activated STAT1, poor consistency was observed from one experiment to another with freshly isolated CD4SP thymocytes. Therefore, we chose to treat the cells with anti-CD3 and anti-CD28 for 3 days prior to the assay. Under such conditions, STAT1 was found to be markedly accumulated in the promoter and regulatory regions of both *Tbx21* and *Ifng* loci of cKO mice (**Figure 5C**). Notably, there were strong positive correlations between the levels of STAT1 binding and H3K4me3 (Pearson's correlation coefficient for *Ifng* locus $r = 0.917$, $p = 0.010$ and *Tbx21* locus $r = 0.991$, $p < 0.001$, **Figure 5D**), indicating the implication of STAT1 in the epigenetic modifications of the loci. Together, these data suggest that IL-27p28-medaited signal normally imposes an inhibitory effect on STAT1 activation, which in turn helps the establishment of functional bias against the Th1 lineage.

6. Augmented autoimmune phenotype of *Aire*-deficient mice in the absence of IL-27p28

The physiological significance remains elusive of the bias against the Th1 effector functions for CD4SP thymocytes and RTEs. One hypothesis proposes that it favors the development of peripheral tolerance by avoiding a more inflammatory and potential detrimental response to self-antigens (Cunningham, Helm, & Fink, 2018; Fink & Hendricks, 2011; Hendricks & Fink, 2011). The disruption of such a bias in *Cd11c-p28$^{f/f}$* mice provided a good model to test this hypothesis. Flow cytometric analysis of the peripheral T cell compartment demonstrated a significant increase of $CD44^{hi}CD62L^{-}$ activated T cells with a concomitant decrease of $CD44^{lo}CD62L^{+}$ naive T cells in the cKO mice compared to the WT littermates (**Figure 6A**). Nevertheless, the cKO mice displayed no obvious signs of autoimmunity up to the age of 24-30 weeks when assessed by auto-antibodies against double-strand DNA (**Figure 6B**) and tissue pathology (**Figure 6C**). Occasionally, small lymphoid foci were observed in the lung of cKO mice. But the $CD4^+$ T cells recovered from the lung tissues displayed a similar cytokine secretion profile to that of WT mice except for the intrinsic augmentation of IFN-γ production (**Figure 6D**). Therefore, IL-27p28 deficiency alone is insufficient to drive the development of autoimmunity.

We next sought to determine whether IL-27p28 deficiency would affect the autoimmune phenotype in the *Aire$^{-/-}$* mouse, which are predisposed to developing autoimmunity due to a defect in clonal deletion of autoreactive T cells (van Laar, Hamburg, & Tas, 2022; J. X. Wang et al., 2021). To this end, *Cd11c-p28$^{f/f}$* mice were crossed with *Aire$^{-/-}$* mice to generate double knockout mice. Histological examination revealed increased lymphocyte infiltration and more severe structural distortion in the lung and stomach in double knockout mice in comparison to mice deficient for Aire only (**Figure 6C**). In addition, the double knockout mice exhibited an elevated level of autoantibodies to double-strand DNA, although the difference did not reach a statistical significance (p=0.068) possibly due to the limited sample size (**Figure 6B**). When $CD4^+$ T cells recovered from the lung tissues were analyzed for IFN-γ, IL-4 and IL-17 production, the double knockout mice were virtually indistinguishable from the *Aire$^{-/-}$* mice, except for increased IFN-γ production (**Figure 6D**).

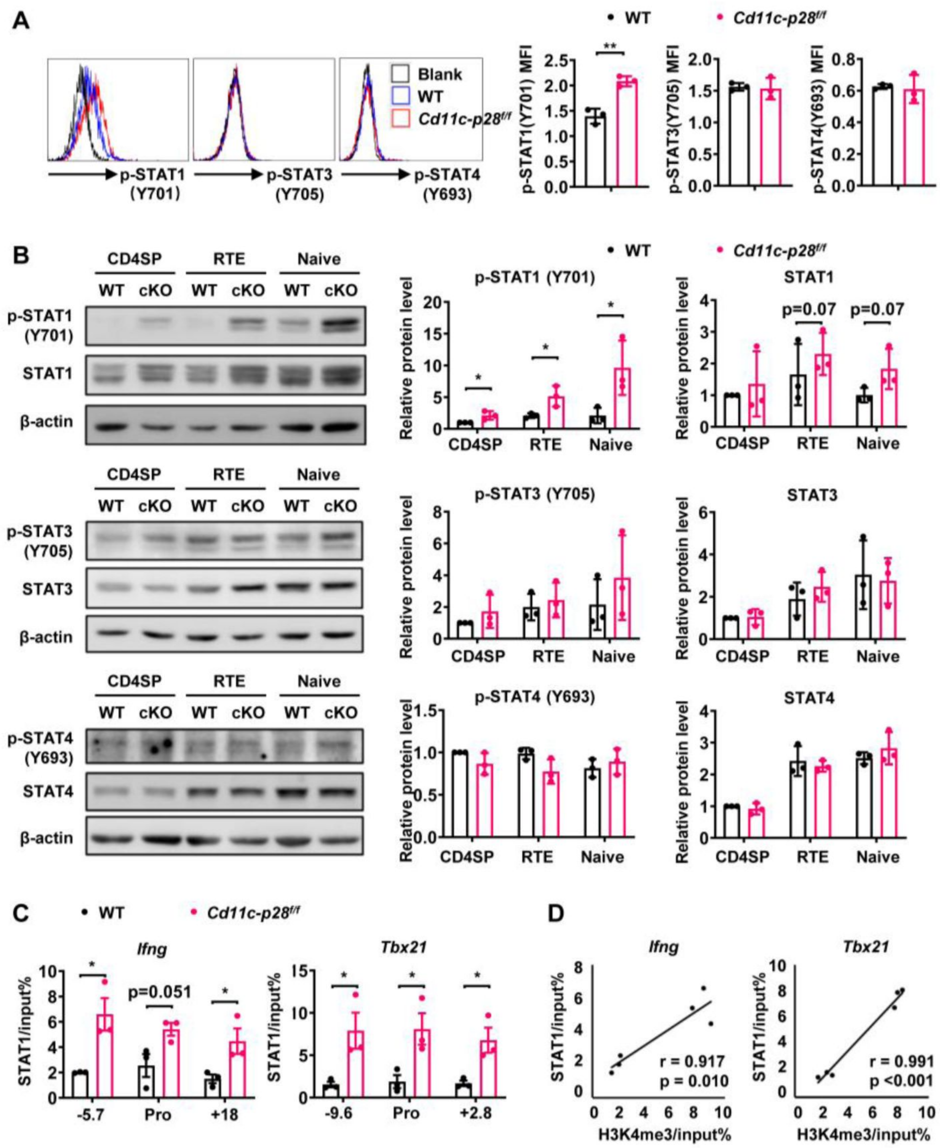


Figure 5.

Enhanced STAT1 activation in CD4SP thymocytes from *Cd11c-p28*$^{f/f}$ mice.

(A) Intracellular staining of freshly isolated thymocytes from *Cd11c-p28*$^{f/f}$ and WT mice using antibodies against phosphorylated STAT1 (Y701), STAT3 (Y705), and STAT4 (Y693). Representative histograms for CD4SP thymocytes (left) and mean fluorescence intensity (MFI) from three independent experiments (right, mean ±SD).

(B) Western blot analysis of total and phosphorylated STAT1 (Y701), STAT3 (Y705), and STAT4 (Y693) in purified CD4SP thymocytes, CD4^{+} RTEs, and naive CD4^{+} T cells from *Cd11c-p28*$^{f/f}$ and WT mice. Representative blots (left) and relative protein levels quantified by densitometry and normalization to β-actin (right, mean ±SD, n=3).

(C) Increased STAT1 binding to promoter and regulatory regions of *Tbx21* and *Ifng* loci in *Cd11c-p28*$^{f/f}$ mice.

(D) Correlation between STAT1 binding and H3K4me3 levels at *Tbx21* and *Ifng* loci. Significance: * *P*<0.05; ** *P*< 0.01.

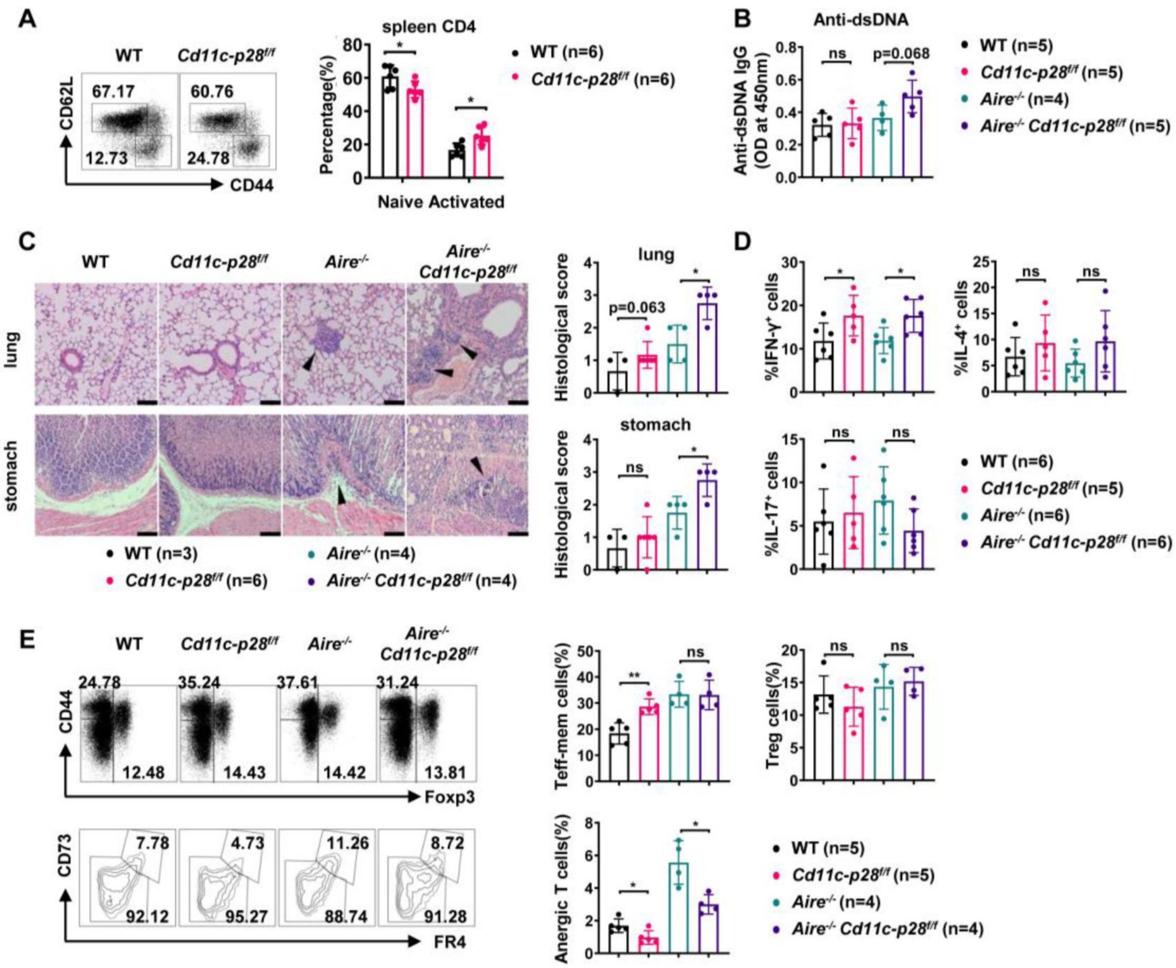


Figure 6.

Exacerbated autoimmune responses in *Aire*$^{-/-}$ mice in the absence of IL27p28.

(A) CD44 and CD62L expression in CD4^{+} T cells from splenocytes of 6-8 week-old WT (n=6) and *Cd11c-p28*$^{f/f}$ (n=6) mice. Representative dot plots (left) and percentages of CD44^{lo}CD62L^{+} naive and CD44^{hi}CD62L^{-} activated T cells are shown (right, mean ±SD).

(B-D) WT, *Cd11c-p28*$^{f/f}$, *Aire*$^{-/-}$, and *Cd11c-p28*$^{f/f}$*Aire*$^{-/-}$ mice (24-30 weeks old) were analyzed. Each symbol represents one mouse (mean ±SD).

(B) Serum anti-dsDNA antibody levels (ELISA).

(C) H&E staining (left) and histological scores (right) of the lung and stomach. Arrows mark lymphocytic infiltrates. Scale bar = 100 μm.

(D) Percentages of IFN-γ^{+}, IL-4^{+}, and IL-17A^{+} CD4^{+} T cells in lung tissue after PMA/ionomycin stimulation.

(E) Splenocytes from 12-week-old mice were stained for CD44, Foxp3, CD73, and FR4. Percentages of anergic (CD4^{+}Foxp3^{-}CD44^{hi}CD73^{hi}FR4^{hi}), effector/memory (CD4^{+}Foxp3^{-}CD44^{hi}CD73^{lo}FR4^{lo}), and regulatory (CD4^{+}Foxp3^{+}) T cells are shown. * *P*<0.05; ** *P*<0.01; ns, not significant.

Immunological anergy constitutes an important mechanism for peripheral tolerance. Two studies have identified a subset of naturally occurring $Foxp3^{-}CD44^{hi}CD73^{hi}FR4^{hi}$ polyclonal $CD4^{+}$ T cells in healthy hosts. They are enriched for self-antigen–specific T cell antigen receptors and represent functionally anergic cells (ElTanbouly & Noelle, 2021; Kalekar et al., 2016). We wondered whether the Th1 bias induced by IL-27p28 deficiency would affect the generation of the anergic population. As shown in **Figure 6E**, the $CD4^{+}Foxp3^{-}CD44^{hi}CD73^{hi}FR4^{hi}$ anergic cells were significantly reduced in the double KO mice, which was accompanied by an increase in the $CD4^{+}Foxp3^{-}CD44^{hi}CD73^{lo}FR4^{lo}$ effector/memory cells. Taken together, these data support that the bias against the Th1 lineage in newly generated T cells provides a unique mechanism to render autoreactive cells exquisitely sensitive to tolerance induction.

Discussion

Th2 polarization in adult SP thymocytes and RTEs is believed to be maintained through epigenetic modification at the *Il4* locus. However, the regulation of the opposing cytokine IFN-γ in this context remains unclear. We reported here that thymic DC-derived IL-27p28 played a key role in the establishment of the repressive status of the *Ifng* locus in newly generated $CD4^{+}$ T cells. As such, the loss of IL-27p28 in DCs endowed these cells with increased capacity of INF-γ production upon TCR engagement. Transcriptome profiling demonstrated coordinated up-regulation of STAT1-activated genes in the absence of IL-27p28. Indeed, enhanced phosphorylation of STAT1 was observed in CD4SP thymocytes from *Cd11c-p28*$^{f/f}$ mice, which was accompanied by the accumulation of STAT1 at the promoter and enhancer regions of *Ifng* and *Tbx21*. Epigenetic analyses indicated reduced DNA methylation at the *Ifng* locus and increased trimethylation of H3K4 at both *Ifng* and *Tbx21* loci in the absence of IL-27p28. Moreover, the H3K4me3 modification was shown to be strongly correlated with STAT1 binding to these loci. These data support a hypothesis that the exposure of developing thymocytes to IL-27p28 induces repressive epigenetic changes at the *Ifng* and *Tbx21* loci, possibly through antagonizing STAT1 activation, ultimately leading to an attenuated IFN-γ response of newly generated CD4SP thymocytes.

The present study revealed a significantly elevated level of phosphorylated STAT1 and up-regulation of a variety of STAT1-regulated genes in CD4SP thymocytes from *Cd11c-p28*$^{f/f}$ mice, indicating that IL-27p28 plays an important role in the fine tuning of STAT1 signaling in the developing thymocytes (Twohig et al., 2019). As IL-27 is known to activate multiple STAT proteins, including STAT1, STAT3, STAT4, and STAT5 following engagement of the gp130/IL-27Rα receptor (Hirahara et al., 2015; Philips et al., 2022), the hyperactivation of STAT1 in the absence of IL-27p28 is somehow unexpected. We examined the potential impact of IL-27 deficiency on the expression of suppressors of cytokine signaling 3 (SOCS3), which has been shown to have a regulatory role in IL-27 signaling (Owaki, Asakawa, Fukai, Mizuguchi, & Yoshimoto, 2006). Comparable levels of SOCS3 were detected in CD4SP thymocytes and naive $CD4^{+}$ T cells from *Cd11c-p28*$^{f/f}$ mice and their littermates (Figure 5-figure supplement 1). We also explored the possibility that the constitutive activation of STAT1 might be the secondary effect of enhanced IFN-γ production (Iwata et al., 2017; Singhania et al., 2019). Addition of IFN-γ antibodies to the culture of CD4SP thymocytes and naive $CD4^{+}$ T cells, however, showed no significant effect on phosphorylated STAT1 levels in knockout cells (Figure 5-figure supplement 2A-B). Thus, STAT1 hyperactivation is unlikely due to loss of SOCS3 expression or positive feedback of IFN-γ signaling. Of note, it has been previously demonstrated that IL-27p28 on its own can act as a natural antagonist of gp130-mediated signaling triggered by IL-6, IL-11 and IL-27 (Stumhofer et al., 2010). Consistent with this finding, Chong et al. reported that IL-27p28 was able to inhibit IL-27-induced Th1 differentiation by reducing the phosphorylation of STAT1 and STAT3 (Chong et al., 2014). Therefore, we speculate that IL-27p28 in the thymus may primarily functions to suppress STAT1 activation induced by other cytokines.

A recent study demonstrated that both recombinant murine and human IL-27 induced the expression of antiviral proteins *Oas1*, *Oas2*, *Oas3*, *Oasl1*, and *Oasl2* in epidermal keratinocytes. IL-27 signaling leads to OAS2 expression in a manner dependent on IL27Rα and STAT1, but independent of STAT2. Apart from their antiviral activity, OAS proteins are also involved in cell growth, differentiation, and apoptosis (Castelli et al., 1997; Huang et al., 2022; Salzberg et al., 1997). The genomic mutation and methylation of most OAS genes have been shown to alter their expression levels, which is associated with the levels of infiltrated $CD4^+$ T cells and $CD8^+$ T cells in the tumor microenvironment (Gao et al., 2022). Secreted OAS2 has also been observed to inhibit CD3 zeta chain expression in T cells (Dar et al., 2016). In our current study, we observed an increase in the transcription of *Oasl2*, *Oas2*, and *Oas3* in CD4SP thymocytes from *Cd11c-p28*$^{f/f}$mice, indicating that IL-27p28 signaling also involved in the fine-tuning of these OAS protein expressions. Whether these molecules play a role in thymocyte development remains to be clarified.

The biased functionality of newly generated $CD4^+$ T cells has been proposed to be important for the induction of peripheral tolerance to self-antigens (Cunningham et al., 2018; Fink & Hendricks, 2011; Hendricks & Fink, 2011). Despite the much enhanced capacity of IFN-γ production by $CD4^+$ T cells and a significant increase of $CD44^{hi}CD62L^-$ activated T cells, *Cd11c-p28*$^{f/f}$ mice displayed no apparent signs of autoimmunity under steady-state conditions. This is consistent with the result of our previous study in which negative selection was found to proceed normally in *Cd11c-p28*$^{f/f}$ mice using the RIP-mOVA and OT-II models (Tang et al., 2016). We further interrogated the impact of the disrupted functional bias in *Aire*$^{-/-}$ mice. Aire induces ectopic expression of self-antigens in medullary thymic epithelial cells to mediate negative selection of autoreactive thymocytes. Its mutation causes a rare life-threatening autoimmune disease affecting multiple organs (van Laar et al., 2022; J. X. Wang et al., 2021). In comparison to *Aire*$^{-/-}$ mice, *Cd11c-p28*$^{f/f}$ *Aire*$^{-/-}$ mice showed elevated levels of double-strand DNA autoantibodies, increased lymphocytes infiltration, and reduced anergic cells. Therefore, although disruption of the functional bias in newly generated T cells is not sufficient to drive autoimmunity, it accelerates disease development in hosts poised for autoimmunity.

In further support of a direct link between IL-27 and Aire, a recent study revealed that IL-27 production was inhibited in Aire-overexpressing murine dendritic cells (Zou et al., 2021). The two subunits IL-27p28 and EBI3 of human IL-27 are usually coordinately expressed in antigen-presenting cells, while mouse IL-27p28 can be secreted independently to inhibit the activities of IL-27 signaling (Pflanz et al., 2002; Stumhofer et al., 2010). The distinct production profile of IL-27p28 between humans and mice and our finding of the critical role of IL-27p28 in the maintenance of functional bias in newly generated $CD4^+$ T cells may explain the much weaker autoimmune phenotype in *Aire*$^{-/-}$ mice than *AIRE*-mutated human subjects. In addition, recent studies have reported elevation of serum IL-27p28 (also known as IL-30) in patients with prostate cancer (Sorrentino et al., 2019), psoriasis (Omar, Xinxin, Jiayi, Hussain, & Biao, 2021), and obesity (Q. Wang et al., 2021). It would be interesting to explore the involvement of altered functionality of newly generated $CD4^+$ T cells under these pathological conditions.

Taken together, our data demonstrate that DC-derived IL-27p28 serves as an endogenous inhibitor for STAT1 hyper-activation in developing thymocytes. By regulating DNA and H3K4 methylation levels at the promoter and transcription regulatory regions of *Ifng* and *Tbx21*, it is critically involved in the establishment of the functional bias against IFN-γ production by newly generated $CD4^+$ T cells. Disruption of this mechanism exacerbates autoimmune phenotypes in hosts predisposed for autoimmunity.

Materials and methods

Mice

C57BL/6 mice were obtained from the Vital River Laboratories (Beijing, China). *Cd11c-p28*$^{f/f}$ mice and *Il27ra*$^{-/-}$ mice on C57BL/6 background were kindly provided by Dr. Zhinan Yin from Jinan University (Guangzhou, China). FVB-Tg (Rag2-EGFP) 1Mnz/J mice were purchased from Jackson Laboratory (Bar Harbor, ME) and were backcrossed for 10 generations onto the C57BL/6 background (termed as RAG2p-EGFP in this paper). RAG2p-EGFP mice were bred with *Cd11c-p28*$^{f/f}$ mice to generate *Cd11c-p28*$^{f/f}$ RAG2p-EGFP mice. *Aire*$^{-/-}$ mice were generously provided by Yangxin Fu (University of Chicago, IL) and were bred with *Cd11c-p28*$^{f/f}$ mice to generate *Aire*$^{-/-}$ *Cd11c-p28* $^{f/f}$ mice. Mice were used at 6-8 weeks of age unless stated otherwise. All the animal procedures were conformed to the Chinese Council on Animal Care Guidelines and the study was approved by the ethics committee of Peking University Health Science Center with an approval number of LA2014178.

Antibodies and reagents

PE-Cy7-conjugated anti-mouse CD4 (RM4-5), PE- and APC-conjugated anti-mouse CD8a (53–6.7), APC-conjugated anti-mouse IL-2 (JES6-5H4), PE-Cy7-conjugated anti-mouse TNF-α (MP6-XT22), PE-conjugated anti-mouse Stat1 (pY701) (4a), PerCP-Cy5.5-conjugated anti-mouse Stat3 (pY705) (4/P-STAT3), Alexa Fluor 488-conjugated anti-mouse Stat4 (pY693) (38/p-Stat4) were purchased from BD Biosciences (San Diego, CA). PE- and APC-conjugated anti-mouse CD25 (PC61.5) and CD44 (IM7), FITC-conjugated anti-mouse FR4 (eBio12A5), PerCP-eFluor710-conjugated anti-mouse CD73 (eBioTY/11.8), FITC-conjugated anti-mouse IL-4 (BVD6-24G2), PE-conjugated anti-mouse IL-17A (eBio17B7), APC-conjugated anti-mouse FOXP3 (3G3), biotin-conjugated anti-mouse CD8a (53–6.7) and anti-phospho-STAT-1(S727) were obtained from eBioscience (Waltham, MA). PE-conjugated anti-mouse IFN-γ (XMG1.2) and PerCP-Cy5.5-conjugated anti-mouse T-bet (4B10) were purchased from BioLegend (San Diego, CA).

Functional grade monoclonal antibodies for murine CD3 (145-2C11) and CD28 (37.51) and neutralizing anti-IL4 (11B11), neutralizing anti-IFN-γ (clone XMG1.2) were obtained from eBioscience (Waltham, MA). Recombinant murine IL-2, IL-4, IL-6, IL-12 and recombinant human TGF-β1 were purchased from R&D Systems (Abingdon, UK).

Anti-phospho-STAT1(Tyr701), anti-STAT1, anti-phospho-STAT3 (Tyr705), anti-STAT3, anti-phospho-STAT4 (Tyr693), anti-STAT4, anti-SOCS3, anti-Actin and HRP-labeled goat-anti-rabbit or anti-mouse IgGs for Western blot assay were purchased from Cell Signaling Technology (Danvers, MA). Phorbol 12-myristate 13-acetate (PMA) and ionomycin were obtained from Sigma-Aldrich.

Cell sorting and CD4$^+$ T-cell differentiation *in vitro*

To enrich CD4SP thymocytes, CD8$^-$ thymocytes were obtained by negative selection using anti-CD8 (Ly-2) MicroBeads (Miltenyi Biotec). The cells were then stained with fluorescently labeled antibodies to CD4, CD8 and CD44. CD4$^+$SP thymocytes with the phenotype of GFP$^+$CD4$^+$CD8$^-$CD44^{lo} were then sorted. For the isolation of CD4$^+$ RTEs and mature naive CD4$^+$ T cells, GFP$^+$CD4$^+$CD8$^-$CD25$^-$NK1.1$^-$ (RTEs) and GFP$^-$CD4$^+$CD8$^-$CD25$^-$CD44^{lo} (naive T) cells were sorted from lymph nodes. All these cells were sorted to > 99% purity with a FACS Aria II (BD Biosciences, San Diego, CA).

For *in vitro* activation, CD4SP thymocytes, RTEs and mature naive T cells were cultured at a density of 2×10^6/mL with plate-coated anti-CD3 (clone 2C11; 2 μg/mL) and soluble anti-CD28 (clone 37N; 1 μg/mL) in RPMl 1640 medium supplemented with 10% fetal bovine serum (Biochrom Ag, Berlin), penicillin, streptomycin and 50 μM 2-mercaptoethanol. Conditions for CD4$^+$ T-cell differentiation were as follows: Th0 cells, IL-2 (2 ng/mL); Th1 cells, IL-12 (10 ng/mL), IL-2 (2 ng/mL), anti-IL-4 (10 μg/mL); Th2 cells, IL-4 (20 ng/mL), IL-2 (2 ng/mL), anti-IFN-γ (10 μg/mL); Th17 cells, recombinant human TGFβ1 (10 ng/mL), IL-6 (10 ng/mL), anti-IL-4 (10 μg/mL), anti-IFN-γ (10 μg/mL); and Treg cells, TGFβ1 (1 ng/mL), IL-2 (10 ng/mL), anti-IL-4 (10 μg/mL), anti-IFN-γ (10 μg/mL). After 3 days of differentiation, supernatants were collected and cells were subjected to intracellular cytokine staining and flow cytometry analyses.

Intracellular staining and flow cytometry

For detection of surface molecules, T cells were labelled with the appropriate fluorescent mAbs on ice for 30 min. For detection of cytoplasmic molecules, T-bet, and FOXP3, T cells were collected, stained with surface molecules, fixed, permeabilized with the Foxp3/Transcription Factor Staining Buffer (eBioscience), and stained with fluorochrome-conjugated mouse antibodies on ice for 30 min. For intracellular cytokine staining, five hours before harvest, T cells were stimulated with PMA (50 ng/mL) plus ionomycin (1 mg/mL) in the presence of a protein transport inhibitor, monensin (2 μM) or BFA (3 μg/mL) (eBioscience). Cells were collected, washed, fixed, permeabilized (FIX AND PERM, Invitrogen) and stained with IFN-γ, IL-4, IL-2, TNF-α, and IL-17 antibodies according to the manufacturer's instructions. For detection of phosphorylated STATs, 1×10^7 thymocytes were fixed for 10 min at 37°C with 2% (wt/vol) paraformaldehyde. After fixing, cells were permeabilized for 30 min on ice with 90% (vol/vol) methanol, and stained with the appropriate antibodies. Flow cytometry was conducted on a Galios (Beckman Coulter) and data analysis was performed using Kaluza software.

Enzyme-Linked Immunosorbent Assay (ELISA)

Supernatants of *in vitro* cell cultures were obtained at 72 hr. IFN-γ and IL-4 production were determined using ELISA kits (BioLegend, San Diego, CA) according to the manufacturer's instructions.

For the analysis of anti-dsDNA antibodies, serum was collected from 24-30-week-old *Aire*$^{-/-}$*Cd11c-Cre p28*$^{f/f}$, *Aire*$^{-/-}$, *Cd11c-Cre p28*$^{f/f}$and WT mice. Nunc MaxiSorp ELISA plates were precoated with dsDNA (100 mg/mL, Sigma, St. Louis, MO) in phosphate-buffered saline (PBS) at 4°C overnight. Plates were blocked with 5% BSA for 1 hour at 37°C, then washed and incubated with 1/100 dilutions of mouse sera for 2 hours at 37°C. Plates were washed, and anti-dsDNA antibodies were detected with a 1/1000 dilution of alkaline phosphatase–conjugated goat anti-mouse IgG (BioLegend) for 1 hour at 37°C and developed with a phosphatase substrate for 30 min at 37°C.

Quantitative PCR (qPCR)

RNA was purified from various T cell subsets cultured under Th0 conditions for 12 h using Trizol reagent (Invitrogen). cDNA was synthesized using reverse transcription kit (Progema). qPCR was performed using FastStart Universal SYBR Green Master mix (Roche, Basel, Switzerland) on an iCycler real-time PCR system (Bio-Rad Laboratories, Hemel Hempstead, U.K.), with each sample in triplicate. The quantification was based on delta delta CT calculations and was normalized to β-actin as loading controls. The primers used in the study were listed in Supplementary Table 1.

DNA methylation

Bisulfite modification of genomic DNA from FACS purified cells was performed using the EZ DNA methylation kit (Zymo Research). For methylation analysis on *Ifng* gene promoter CpG sites, bisulfite-treated DNA was amplified in semi-nested PCR using primers: 5'-GGTGTGAAGTAAAAGTGTTTTTAGAGAATTTTAT-3' and 5'-CAATAACAACCAAAAACAACCATAAAAAAAAAACT-3', then 5'-GGTGTGAAGTAAAAGTGTTTTTAGAGAATTTTAT-3' and 5'-CCATAAAAAAAAAACTACAAAACCAAAATACAATA-3'. For methylation analysis on *Il4* gene promoter CpG sites, bisulfite-treated DNA was amplified in PCR using primers: 5-

GGATCCACACGGTGCAAAGAGAGACCC-3’ and 5'-TCGGCCTTTCAGACTAATCTTATCAGC-3’ The PCR products were gel purified and cloned into the pGEM-T vector (Promega; Madison, WI, USA). The inserted PCR fragments of individual clones were sequenced by Tsing KE Biological Technology, Beijing, China. For all samples, 10 reads per CpG site were used to determine the average percentage of methylated CpG.

Chromatin immunoprecipitation (ChIP)

ChIP assays were carried out by using Pierce Agarose Chip Kit (Pierce Biotechnology). Briefly, cells were cross-linked for 10 min with 1% formaldehyde and lysed with Nuclease. 10% of the digested chromatin was preserved as input control and the rest of the digested chromatin were incubated with specific antibody or normal IgG (as a control). The purified DNA from immunoprecipitation and the input samples were analyzed by qPCR using primers specific listed in Supplementary Table 1. Data is presented as a percent input of each IP sample relative to input chromatin. The following antibodies were used in ChIP analysis: H3K4me3 (Millipore), H3K27me3 (Millipore), H3K9me3 (Abcam), and STAT1 (Cell Signaling Technology).

Western blotting

For short term TCR stimulation, CD4SP thymocytes and mature naive T cells were incubated with 2 μg/mL anti-mouse CD3 and 1 μg/mL anti-mouse CD28 on ice for 20 min, followed by cross-linking for 10 min at 37°C with 5 μg/mL goat-anti-hamster IgG. Cells were then washed with PBS. Freshly isolated T cells or stimulated T cells were lysed in RIPA buffer for 30 minutes. Whole-cell lysates (5–10 μg per sample) were separated by SDS/PAGE and analyzed by immunoblotting with antibodies to phospho-STAT-1 (Tyr701), phospho-STAT-1 (Ser727), STAT-1, phospho-STAT-3 (Tyr705), STAT-3, phospho-STAT-4 (Tyr693), STAT-4, SOCS3 and Actin. HRP-conjugated anti-Rabbit IgG was used as the detection antibody. The bands were quantified using ImageJ software.

RNA-seq

Total RNA was extracted from approximately 2×10^6 CD4SP thymocytes from WT and *Cd11c-p28*$^{f/f}$ mice using Trizol reagent (Invitrogen). Each sample was a mixture of equal amounts of CD4SP thymocytes from three mice. RNA underwent quality control testing using a bionalayser followed by cDNA library preparation. Library construction and sequencing on Illumina HiSEq 2000 were performed by BIOPIC, Peking University.

H&E

Organs from 24-30-week-old *Aire*$^{-/-}$*Cd11c-p28*$^{f/f}$ mice, *Aire*$^{-/-}$ mice and WT mice were collected and fixed overnight in 10% formalin, embedded in paraffin, sectioned and stained with hematoxylin and eosin (H&E). The degree of lymphocytic infiltrates was analyzed in a blinded fashion. In general, 0, 1, 2 and 3 indicate no, mild, moderate, or severe lymphocytic infiltration, respectively.

Statistical analysis

Data are reported as the mean ± Standard deviation (SD). Differences between groups were analyzed by Student’s *t* test or two-way Anova (for multiple variant comparisons) using GraphPad Prism software (GraphPad). Throughout the text, figures, and figure legends, the following terminology is used to denote statistical significance: * P<0.05, ** P <0.01. *** P<0.001. NS, not significant.

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Additional information

Author Contributions

JZ and RJ designed and carried out the study, collected and analyzed the data and drafted the manuscript. JZ and HT performed the phenotype analysis, epigenetic assay and established the mouse model. HMW analyzed the RNA-seq data. XWP performed the cell sorting assay. YZ conceived and designed the study, analyzed the data, and drafted the manuscript. All authors read and approved the final manuscript.

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Abbreviations

- AS: ankylosing spondyliti
- BD: behcet disease
- ChIP: chromatin immunoprecipitation
- cKO: conditional knockout
- DC: dendritic cell
- DEGs: differentially expressed genes
- EAE: experimental autoimmune encephalomyelitis
- EBI3: Epstein-Barr virus-induced gene 3
- ELISA: Enzyme-Linked Immunosorbent Assay
- GO: gene ontology
- GSEA: gene set enrichment analysis
- H3K4me3: histone H3 lysine 4 trimethylation
- H3K27me3: histone H3 lysine 27 trimethylation
- H3K9me3: histone H3 lysine 9 trimethylation
- IFN-γ: interferon-γ
- IL-2: interleukin 2
- KEGG: Kyoto Encyclopedia of Genes and Genomes
- PBS: phosphate-buffered saline
- RTE: recent thymic emigrants
- SOCS3: suppressors of cytokine signaling 3
- SP: single-positive thymocyte
- STAT: signal transducer and activator of transcription
- T-bet: T-box transcription factor
- Th: T helper
- TNF-α: tumor necrosis factor α
- WT: wild-type

Additional files

Supplementary material.

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Reviewer #1 (Public Review):

Summary:

Zhang et al. demonstrate that CD4+ single positive (SP) thymocytes, CD4+ recent thymic emigrants (RTE), and CD4+ T naive (Tn) cells from Cd11c-p28-flox mice, which lack IL-27p28 selectively in Cd11c+ cells, exhibit a hyper-Th1 phenotype instead of the expected hyper Th2 phenotype. Using IL-27R-deficient mice, the authors confirm that this hyper-Th1 phenotype is due to IL-27 signaling via IL-27R, rather than the effects of monomeric IL-27p28. They also crossed Cd11c-p28-flox mice with autoimmune-prone Aire-deficient mice and showed that both T cell responses and tissue pathology are enhanced, suggesting that SP, RTE, and Tn cells from Cd11c-p28-flox mice are poised to become Th1 cells in response to self-antigens. Regarding mechanism, the authors demonstrate that SP, RTE, and Tn cells from Cd11c-p28-flox mice have reduced DNA methylation at the IFN-g and Tbx21 loci, indicating 'de-repression', along with enhanced histone tri-methylation at H3K4, indicating a 'permissive' transcriptional state. They also find evidence for enhanced STAT1 activity, which is relevant given the well-established role of STAT1 in promoting Th1 responses, and surprising given IL-27 is a potent STAT1 activator. This latter finding suggests that the Th1-inhibiting property of thymic IL-27 may not be due to direct effects on the T cells themselves.

Strengths:

Overall the data presented are high quality and the manuscript is well-reasoned and composed. The basic finding - that thymic IL-27 production limits the Th1 potential of SP, RTE, and Tn cells - is both unexpected and well described.

Weaknesses from the original round of review:

A credible mechanistic explanation, cellular or molecular, is lacking. The authors convincingly affirm the hyper-Th1 phenotype at epigenetic level but it remains unclear whether the observed changes reflect the capacity of IL-27 to directly elicit epigenetic remodeling in developing thymocytes or knock-on effects from other cell types which, in turn, elicit the epigenetic changes (presumably via cytokines). The authors propose that increased STAT1 activity is a driving force for the epigenetic changes and resultant hyper-Th1 phenotype. That conclusion is logical given the data at hand but the alternative hypothesis - that the hyper-STAT1 response is just a downstream consequence of the hyper-Th1 phenotype - remains equally likely. Thus, while the discovery of a new anti-inflammatory function for IL-27 within the thymus is compelling, further mechanistic studies are needed to advance the finding beyond phenomenology.

https://doi.org/10.7554/eLife.96868.2.sa1

Reviewer #2 (Public Review):

Summary:

Naïve CD4 T cells in CD11c-Cre p28-floxed mice express highly elevated levels of proinflammatory IFNg and the transcription factor T-bet. This phenotype turned out to be imposed by thymic dendritic cells (DCs) during CD4SP T cell development in the thymus [PMID: 23175475]. The current study affirms these observations, first, by developmentally mapping the IFNg dysregulation to newly generated thymic CD4SP cells [PMID: 23175475], second, by demonstrating increased STAT1 activation being associated with increased T-bet expression in CD11c-Cre p28-floxed CD4 T cells [PMID: 36109504], and lastly, by confirming IL-27 as the key cytokine in this process [PMID: 27469302]. The authors further demonstrate that such dysregulated cytokine expression is specific to the Th1 cytokine IFNg, without affecting the expression of the Th2 cytokine IL-4, thus proposing a role for thymic DC-derived p28 in shaping the cytokine response of newly generated CD4 helper T cells. Mechanistically, CD4SP cells of CD11c-Cre p28-floxed mice were found to display epigenetic changes in the Ifng and Tbx21 gene loci that were consistent with increased transcriptional activities of IFNg and T-bet mRNA expression. Moreover, in autoimmune Aire-deficiency settings, CD11c-Cre p28-floxed CD4 T cells still expressed significantly increased amounts of IFNg, exacerbating the autoimmune response and disease severity. Based on these results, the investigators propose a model where thymic DC-derived IL-27 is necessary to suppress IFNg expression by CD4SP cells and thus would impose a Th2-skewed predisposition of newly generated CD4 T cells in the thymus, potentially relevant in autoimmunity.

Strengths:

Experiments are well-designed and executed. The conclusions are convincing and supported by the experimental results.

Weaknesses from the original round of review:

The premise of the current study is confusing as it tries to use the CD11c-p28 floxed mouse model to explain the Th2-prone immune profile of newly generated CD4SP thymocytes. Instead, it would be more helpful to (1) give full credit to the original study which already described the proinflammatory IFNg+ phenotype of CD4 T cells in CD11c-p28 floxed mice to be mediated by thymic dendritic cells [PMID: 23175475], and then, (2) build on that to explain that this study is aimed to understand the molecular basis of the original finding.

In its essence, this study mostly rediscovers and reaffirms previously reported findings, but with different tools. While the mapping of epigenetic changes in the IFNg and T-bet gene loci and the STAT1 gene signature in CD4SP cells are interesting, these are expected results, and they only reaffirm what would be assumed from the literature. Thus, there is only incremental gain in new insights and information on the role of DC-derived IL-27 in driving the Th1 phenotype of CD4SP cells in CD11c-p28 floxed mice.

Altogether, the major issues of this study remain unresolved:

(1) It is still unclear why the p28-deficiency in thymic dendritic cells would result in increased STAT1 activation in CD4SP cells. Based on their in vitro experiments with blocking anti-IFNg antibodies, the authors conclude that it is unlikely that the constitutive activation of STAT1 would be a secondary effect due to autocrine IFNg production by CD4SP cells. However, this possibility should be further tested with in vivo models, such as Ifng-deficient CD11c-p28 floxed mice. Alternatively, is this an indirect effect by other IFNg producers in the thymus, such as iNKT cells? It is necessary to explain what drives the STAT1 activation in CD11c-p28 floxed CD4SP cells in the first place.

(2) It is also unclear whether CD4SP cells are the direct targets of IL-27 p28. The cell-intrinsic effects of IL-27 p28 signaling in CD4SP cells should be assessed and demonstrated, ideally by CD4SP-specific deletion of IL-27Ra, or by establishing bone marrow chimeras of IL-27Ra germline KO mice.

[Editors' note: The resubmitted paper was minimally revised, and many of the initial concerns remain unresolved.]

https://doi.org/10.7554/eLife.96868.2.sa0

Author response:

The following is the authors' response to the original reviews

> ***Public Reviews:***
>
> ***Reviewer #1 (Public Review):***
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> *Summary:*
>
> *Zhang et al. demonstrate that CD4$^+$ single positive (SP) thymocytes, CD4$^+$ recent thymic emigrants (RTE), and CD4$^+$ T naive (Tn) cells from Cd11c-p28-flox mice, which lack IL-27p28 selectively in Cd11c+ cells, exhibit a hyper-Th1 phenotype instead of the expected hyper Th2 phenotype. Using IL-27R-deficient mice, the authors confirm that this hyper-Th1 phenotype is due to IL-27 signaling via IL-27R, rather than the effects of monomeric IL-27p28. They also crossed Cd11c-p28-flox mice with autoimmune-prone Aire-deficient mice and showed that both T cell responses and tissue pathology are enhanced, suggesting that SP, RTE, and Tn cells from Cd11c-p28-flox mice are poised to become Th1 cells in response to self-antigens. Regarding mechanism, the authors demonstrate that SP, RTE, and Tn cells from Cd11c-p28-flox mice have reduced DNA methylation at the IFN-g and Tbx21 loci, indicating 'de-repression', along with enhanced histone tri-methylation at H3K4, indicating a 'permissive' transcriptional state. They also find evidence for enhanced STAT1 activity, which is relevant given the well-established role of STAT1 in promoting Th1 responses, and surprising given IL-27 is a potent STAT1 activator. This*

latter finding suggests that the Th1-inhibiting property of thymic IL-27 may not be due to direct effects on the T cells themselves.

Strengths:

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Weaknesses:

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Thank you for your insightful comments and suggestions. We appreciate your feedback and have carefully considered the concerns raised regarding the mechanistic explanation of our findings. To address the issue of whether developing thymocytes are the direct targets of IL-27, we plan to conduct further studies using *Cd4-IL-27ra* knockout mice or mixed bone marrow chimeras consisting of wildtype and *IL-27ra* knockout cells. This approach will help us determine if IL-27 directly induces epigenetic remodeling in thymocytes or if the observed effects are secondary to influences from other cell types.

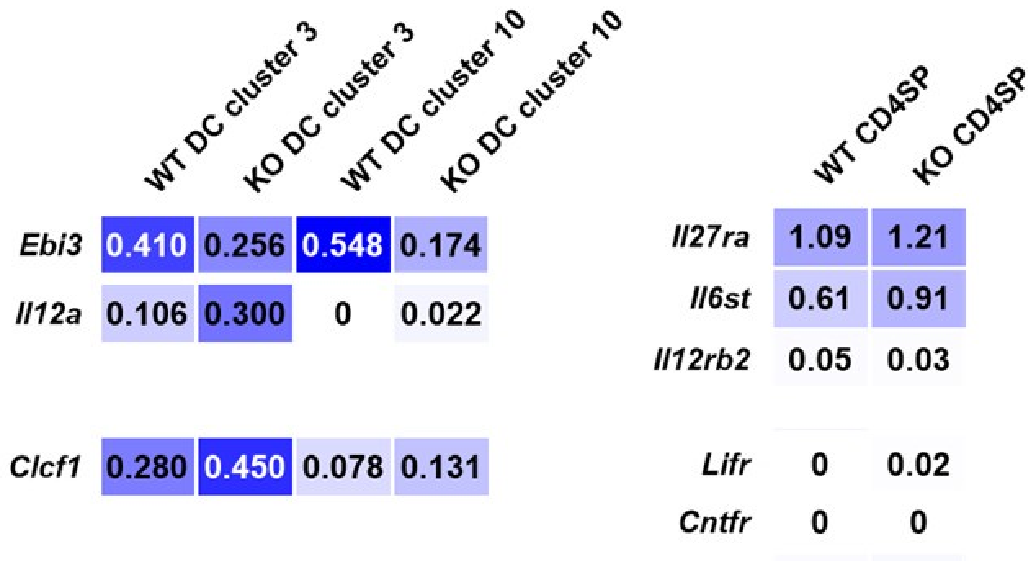
Regarding the potential autocrine loop contributing to STAT1 hyperactivation, we have performed preliminary experiments by adding IFN-γ antibody to $CD4^+$ T cell cultures and observed no significant impact on STAT1 phosphorylation. If necessary, we will further investigate this possibility in vivo using *Cd4-Ifng* and *CD11c-p28* double knockout mice.

The detailed mechanisms underlying STAT1 hyperactivation remain to be elucidated. Recent studies have shown that IL-27p28 can act as an antagonist of gp130-mediated signaling. Structural analyses have also demonstrated that IL-27p28 interacts with EBI3 and the two receptor subunits IL-27Rα and gp130. Given these findings and the similar phenotypes observed in p28 and IL-27ra deficient mice, we speculate that the deficiency of either p28 or IL-27ra may increase the availability of gp130 for signaling by other cytokines. We will focus our future research on gp130-related cytokines to identify potential candidates that could lead to enhanced STAT1 activation in the absence of p28. Alternatively, the release of EBI3 in p28-deficient conditions may promote its interaction with other cytokine subunits. IL-35, which is composed of EBI3 and p35, is of particular interest given the involvement of IL-27Rα in its signaling pathway.

To narrow down the candidate cytokines, we reanalyzed single-cell RNA sequencing data from *CD11c-cre* $p28^{f/f}$ and wild-type thymocytes (Signal Transduct Target Ther. 2022, DOI: 10.1038/s41392-022-01147-z). Our analysis revealed that thymic dendritic cells (DCs) were categorized into two distinct clusters, with both *Il12a* (p35, which forms IL-35 with EBI3) and *Clcf1* (CLCF1) being upregulated in *CD11c-cre* $p28^{f/f}$ mice. In CD4 single-positive (SP) thymocytes, the expression levels of gp130 and IL-12Rβ2 (the receptor for IL-35) were comparable between knockout and wild-type mice. However, the mRNA levels of *Lifr* and *Cntfr* were low in CD4 SP thymocytes.

Author response image 1.

Single-cell RNA sequencing data from CD11c-cre *p28*$^{f/f}$ (KO) and wild-type thymocytes (Signal Transduct Target Ther. 2022, DOI: 10.1038/s41392-022-01147-z).



We have planned to assess the protein levels of IL-35 and CLCF1 in dendritic cells, as well as their respective receptors, to evaluate their effects on STAT1 phosphorylation in CD4^{+} thymocytes from both wild-type and p28-deficient mice. Unfortunately, we have encountered challenges with the mouse breeding and anticipate that it will take approximately six months to obtain the appropriate genotype necessary to complete these experiments.

Reviewer #2 (Public Review):

Summary:

Naïve CD4 T cells in CD11c-Cre p28-floxed mice express highly elevated levels of proinflammatory IFNg and the transcription factor T-bet. This phenotype turned out to be imposed by thymic dendritic cells (DCs) during CD4SP T cell development in the thymus [PMID: 23175475]. The current study affirms these observations, first, by developmentally mapping the IFNg dysregulation to newly generated thymic CD4SP cells [PMID: 23175475], second, by demonstrating increased STAT1 activation being associated with increased T-bet expression in CD11c-Cre p28-floxed CD4 T cells [PMID: 36109504], and lastly, by confirming IL-27 as the key cytokine in this process [PMID: 27469302]. The authors further demonstrate that such dysregulated cytokine expression is specific to the Th1 cytokine IFNg, without affecting the expression of the Th2 cytokine IL-4, thus proposing a role for thymic DC-derived p28 in shaping the cytokine response of newly generated CD4 helper T cells. Mechanistically, CD4SP cells of CD11c-Cre p28-floxed mice were found to display epigenetic changes in the Ifng and Tbx21 gene loci that were consistent with increased transcriptional activities of IFNg and T-bet mRNA expression. Moreover, in autoimmune Aire-deficiency settings, CD11c-Cre p28-floxed CD4 T cells still expressed significantly increased amounts of IFNg, exacerbating the autoimmune response and disease severity. Based on these results, the investigators propose a model where thymic DC-derived IL-27 is necessary to suppress IFNg expression by CD4SP cells and thus would impose a Th2-skewed predisposition of newly generated CD4 T cells in the thymus, potentially relevant in autoimmunity.

Strengths:

Experiments are well-designed and executed. The conclusions are convincing and supported by the experimental results.

Weaknesses:

The premise of the current study is confusing as it tries to use the CD11c-p28 floxed mouse model to explain the Th2-prone immune profile of newly generated CD4SP thymocytes. Instead, it would be more helpful to (1) give full credit to the original study which already described the proinflammatory IFNg+ phenotype of CD4 T cells in CD11c-p28 floxed mice to be mediated by thymic dendritic cells [PMID: 23175475], and then, (2) build on that to explain that this study is aimed to understand the molecular basis of the original finding.

In its essence, this study mostly rediscovers and reaffirms previously reported findings, but with different tools. While the mapping of epigenetic changes in the IFNg and T-bet gene loci and the STAT1 gene signature in CD4SP cells are interesting, these are expected results, and they only reaffirm what would be assumed from the literature. Thus, there is only incremental gain in new insights and information on the role of DC-derived IL-27 in driving the Th1 phenotype of CD4SP cells in CD11c-p28 floxed mice.

Thank you for your valuable comments and suggestions. We appreciate your input and have carefully reviewed the concerns raised regarding the premise and novelty of our study.

Indeed, the current study is built upon the foundational work of Zhang et al. (PMID: 23175475), which first described the proinflammatory IFN-γ^{+} phenotype of CD4 T cells in CD11c-p28 floxed mice mediated by thymic dendritic cells. We have cited this study multiple times in our manuscript to acknowledge its significance. Our goal was to expand on this original finding by exploring the functional bias of newly generated $CD4^{+}$ T cells, elucidating the mechanisms underlying the hyper-Th1 phenotype in the absence of thymic DC-derived IL-27, and exploring its relevance in pathogenesis of autoimmunity.

Our study revisits this phenomenon with a focus on the molecular and epigenetic changes that drive the Th1 bias in CD4SP cells. We demonstrated that the deletion of p28 in thymic dendritic cells leads to an unexpected hyperactivation of STAT1, which is associated with epigenetic modifications that favor Th1 differentiation. These findings provide a deeper understanding of the molecular basis behind the original observation of the Th1-skewed phenotype in CD11c-p28 floxed mice.

However, as you pointed out, there is still a gap in understanding the precise link between p28 deficiency and STAT1 activation. We acknowledge that our study primarily reaffirms previously reported findings with different tools and approaches. While the mapping of epigenetic changes in the IFN-γ and T-bet gene loci and the STAT1 gene signature in CD4SP cells are interesting, they are indeed expected results based on the existing literature. This limits the novelty and incremental gain in new insights provided by our study.

To address this gap and enhance the novelty of our findings, we plan to conduct further investigations to elucidate the detailed mechanisms connecting p28 deficiency to STAT1 hyperactivation. We will explore potential compensatory pathways or alternative signaling mechanisms that may contribute to the observed epigenetic changes and Th1 bias. Additionally, we will consider the broader impact of IL-27 deficiency on the thymic environment and its downstream effects on $CD4^{+}$ T cell differentiation.

We appreciate your feedback and will work to strengthen the mechanistic underpinnings of our study. We believe that these additional efforts will provide a more comprehensive understanding of the role of DC-derived IL-27 in shaping the Th1 phenotype of CD4SP cells and contribute meaningful insights to the field.

> *Altogether, the major issues of this study remain unresolved:*
>
> *(1) It is still unclear why the p28-deficiency in thymic dendritic cells would result in increased STAT1 activation in CD4SP cells. Based on their in vitro experiments with blocking anti-IFNg antibodies, the authors conclude that it is unlikely that the constitutive activation of STAT1 would be a secondary effect due to autocrine IFNg production by CD4SP cells. However, this possibility should be further tested with in vivo models, such as Ifng-deficient CD11c-p28 floxed mice. Alternatively, is this an indirect effect by other IFNg producers in the thymus, such as iNKT cells? It is necessary to explain what drives the STAT1 activation in CD11c-p28 floxed CD4SP cells in the first place.*

Thank you for your insightful suggestions. We appreciate your feedback and are committed to addressing the critical questions raised regarding the mechanisms underlying STAT1 activation in CD4SP cells in the context of p28 deficiency in thymic dendritic cells.

To further investigate the potential autocrine loop for IFN-γ production, we will conduct in vivo studies using Cd4-Ifng and CD11c-p28 double knockout mice. This model will allow us to directly test whether IFN-γ produced by CD4SP cells themselves contributes to the observed STAT1 activation. Additionally, this approach will help exclude the possibility of indirect effects from other IFN-γ-producing cells in the thymus, such as invariant natural killer T (iNKT) cells, as suggested by the reviewer.

As you correctly pointed out, a key unanswered question is what drives the initial STAT1 activation in CD4SP cells of CD11c-p28 floxed mice. Our current hypothesis is that p28 deficiency enhances the responsiveness of developing thymocytes to STAT1-activating cytokines. This hypothesis is supported by several lines of evidence:

(1) Functional Antagonism: Recent studies have shown that IL-27p28 can act as an antagonist of gp130-mediated signaling. This suggests that in the absence of p28, the inhibitory effect of IL-27p28 on downstream signaling may be lost, leading to increased sensitivity to other cytokines that activate STAT1.

(2) Structural Insights: Structural studies have demonstrated that IL-27p28 is centrally positioned within the complex formed with EBI3 and the two receptor subunits IL-27Rα and gp130. This positioning implies that p28 deficiency could disrupt the balance of cytokine signaling pathways involving these components.

(3) Phenotypic Similarity: We have observed a similar hyper-Th1 phenotype in mice lacking either p28 or IL-27ra. This similarity suggests that the absence of p28 may lead to increased availability of gp130 for signaling by other cytokines, thereby enhancing STAT1 activation.

Based on these considerations, we hypothesize that the deficiency of p28 results in a greater availability of gp130 to transduce signals from other cytokines, ultimately leading to enhanced STAT1 activation in CD4SP cells. To identify the specific cytokine(s) responsible for this effect, we will focus on gp130-related cytokines, as outlined in our response to Reviewer 1. This will involve reanalysis of single-cell RNA sequencing data and further experimental validation to pinpoint the candidate cytokines driving the observed STAT1 hyperactivation.

We are confident that these additional studies will provide a clearer understanding of the mechanisms linking p28 deficiency in thymic dendritic cells to increased STAT1 activation in CD4SP cells. We appreciate your guidance and look forward to sharing our findings.

> *(2) It is also unclear whether CD4SP cells are the direct targets of IL-27 p28. The cell-intrinsic effects of IL-27 p28 signaling in CD4SP cells should be assessed and*

> *demonstrated, ideally by CD4SP-specific deletion of IL-27Ra, or by establishing bone marrow chimeras of IL-27Ra germline KO mice.*

Thanks for the suggestions. Further studies will be performed to test whether developing thymocytes are the direct targets of IL-27 using *Cd4-IL-27ra* knockout mice or mixed bone marrow chimeras of wildtype and *IL-27ra* knockout cells. Unfortunately, we have encountered challenges with the mouse breeding and anticipate that it will take approximately six months to obtain the appropriate genotype necessary to complete these experiments.

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations For The Authors):***
>
> *(1) Is the hyper-STAT1 response seen in T cells from Cd11c-p28-flox mice due to increased availability and/or increased responsiveness to STAT1 activating cytokines? Studies, where SP, RTE, and Tn cells are pulsed ex vivo with IL-27 and/or other STAT1-activating cytokines, would address the latter (with STAT1 phosphorylation as the major readout). Given the ability of IL-27 to activate STAT3, this pathway should also be addressed. It would be of interest if STAT1 signaling is selectively impaired, as suggested by the work of Twohig et al. (doi: 10.1038/s41590-019-0350-0.)(which should be cited and discussed).*

Thank you for your insightful suggestions. We appreciate your input and are committed to addressing the critical questions raised regarding the mechanisms underlying the hyper-activation of STAT1 in T cells from Cd11c-p28-flox mice.

The detailed mechanisms driving the hyper-activation of STAT1 remain to be fully elucidated. Recent studies have shown that IL-27p28 can act as an antagonist of gp130-mediated signaling. Structural analyses have also demonstrated that IL-27p28 interacts with EBI3 and the two receptor subunits IL-27Rα and gp130. Considering these findings and the similar phenotypes observed in p28 and IL-27ra deficient mice, we speculate that the deficiency of either p28 or IL-27ra may increase the availability of gp130 for signaling by other cytokines. This could potentially enhance the responsiveness of developing thymocytes to STAT1-activating cytokines. We will focus our future research on gp130-related cytokines to identify the candidate(s) responsible for the enhanced STAT1 activation in the absence of p28. Alternatively, the release of EBI3 in the absence of p28 may facilitate its coupling with other cytokine subunits. IL-35, which is composed of EBI3 and p35, is of particular interest given the involvement of IL-27Rα in its signaling pathway.

To narrow down the candidate cytokines, we reanalyzed single-cell RNA sequencing data from CD11c-cre *p28*$^{f/f}$ and wild-type thymocytes (Signal Transduct Target Ther. 2022, DOI: 10.1038/s41392-022-01147-z). Our analysis revealed that thymic dendritic cells (DCs) were categorized into two distinct clusters, with both *Il12a* (p35, which forms IL-35 with EBI3) and *Clcf1* (CLCF1) being upregulated in *CD11c-cre p28*$^{f/f}$ mice. In CD4 single-positive (SP) thymocytes, the expression levels of gp130 and IL-12Rβ2 (the receptor for IL-35) were comparable between knockout and wild-type mice. However, the mRNA levels of *Lifr* and *Cntfr* were low in CD4 SP thymocytes.

Single-cell RNA sequencing data from CD11c-cre *p28*$^{f/f}$ (KO) and wild-type thymocytes (Signal Transduct Target Ther. 2022, DOI: 10.1038/s41392-022-01147-z).

We have planned to assess the protein levels of IL-35 and CLCF1 in dendritic cells, as well as their respective receptors, to evaluate their effects on STAT1 phosphorylation in CD4$^+$ thymocytes from both wild-type and p28-deficient mice. Unfortunately, we have encountered challenges with the mouse crosses and anticipate that it will take approximately six months to obtain the appropriate genotype necessary to complete these experiments.

As you correctly noted, the ability of IL-27 to activate STAT3 signaling is an important consideration. We have carefully examined this pathway in our current study, and our results indicate that neither total nor phosphorylated STAT3 and STAT4 were found to be altered with IL-27p28 ablation (Figure 5B). This suggests that the impact is indeed specific to the STAT1 axis. We will also consider the possibility of selective impairment of STAT1 signaling, as suggested by the work of Twohig et al. (doi: 10.1038/s41590-019-0350-0), which we will cite and discuss in our revised manuscript.

We appreciate your guidance and will work diligently to address these questions in our future studies. We look forward to sharing our findings and contributing to a deeper understanding of the role of IL-27 in the regulation of STAT1 activation in T cells.

> *(2) It may be that the hyper-Th1 phenotype is not due to cell-intrinsic differences in STAT1 signaling (see Major Point 1) but rather, hyper-responsiveness to TCR + Co-stimulation (as provided in the re-stim assays used throughout). This issue is particularly relevant for the ChIP studies where the author notes that, "...we chose to treat the cells with anti-CD3 and anti-CD28 for 3 days prior to the assay". Why not treat these cells ex vivo with STAT1-activating cytokines instead of anti-CD3/CD28? The current methodology makes it impossible to distinguish between enhanced TCR/CD28 and cytokine signaling, and ultimately does not address SP, RTE, and Tn cells (since they are now activated, blasts.).*

Thank you for raising this important point. We appreciate your feedback and fully recognize the limitations of our current methodology, which uses anti-CD3/CD28 stimulation for ChIP experiments. This approach indeed complicates the distinction between enhanced TCR/CD28 signaling and cytokine-mediated STAT1 activation, particularly in the context of SP, RTE, and Tn cells, which become activated blasts under these conditions.

To address these concerns and provide more precise insights into the mechanisms underlying the hyper-Th1 phenotype, we are revising our experimental strategy. Specifically, we are shifting our focus to directly investigate the role of STAT1-activating cytokines in the absence of p28. Based on our previous analysis and re-evaluation of single-cell RNA sequencing data, we have identified IL-35 and CLCF1 as the most promising candidate cytokines.

We are now planning to perform ChIP experiments using these cytokines directly, rather than relying on TCR + co-stimulation. This approach will allow us to more accurately evaluate the impact of these cytokines on STAT1 signaling in $CD4^+$ T cells. By treating cells *ex vivo* with IL-35 and CLCF1, we aim to elucidate whether the observed hyper-Th1 phenotype is driven by enhanced responsiveness to these cytokines, independent of TCR/CD28 signaling.

We regret to inform you that we have encountered unforeseen challenges with mouse crosses, which have delayed our progress. As a result, we anticipate a delay of approximately six months to obtain the appropriate genotypes necessary to complete these experiments. We understand the importance of these revisions and are committed to overcoming these challenges to provide a more robust and accurate analysis.

> *(3) Studies involving STAT1-deficient mice are necessary (ideally with STAT1 deficiency restricted to the T cell compartment). At a minimum, it must be confirmed that these phenocopy Cd11c-p28-flox mice in terms of SP, RTE, and Tn cells (and their Th1-like character). If a similar hyper-Th1 phenotype is not seen, then the attendant hyper STAT1 response can only be viewed as a red herring.*

Thank you for raising this important consideration. We acknowledge the significance of addressing the role of STAT1 specifically within the T cell compartment to validate the mechanisms underlying the hyper-Th1 phenotype observed in Cd11c-p28-flox mice.

We agree that studies involving STAT1-deficient mice, particularly with STAT1 deficiency restricted to the T cell compartment, are essential to confirm whether the hyper-Th1 phenotype is directly driven by STAT1 hyperactivation in T cells. Ideally, such studies would help determine if STAT1 deficiency in T cells phenocopies the Cd11c-p28-flox mice, particularly in terms of the SP, RTE, and Tn cells and their Th1-like characteristics.

Unfortunately, we currently face challenges in obtaining and breeding the appropriate STAT1 conditional knockout mice with T cell-specific deletion. This has limited our ability to conduct these experiments in a timely manner. However, we recognize the importance of these studies and are actively working to secure the necessary resources and models to address this critical question.

We understand that without these experiments, any conclusions drawn about the role of STAT1 hyperactivation in driving the hyper-Th1 phenotype must be considered with caution. If a similar hyper-Th1 phenotype is not observed in STAT1-deficient T cells, then the hyper-STAT1 response may indeed be a secondary or compensatory effect rather than a primary driver.

We are committed to pursuing these studies and will prioritize them in our future work. We will keep you informed of our progress and will update the manuscript with the results of these experiments once completed. We appreciate your patience and understanding as we work to address this important aspect of our research.

> *(4) The authors mine their RNA-seq data using a STAT1 geneset sourced from studies involving IL-21 as the upstream stimulus. Why was this geneset was chosen? It is true that IL-21 can activate STAT1 but STAT3 is typically viewed as its principal signaling pathway. There are many more appropriate genesets, especially from studies where T cells are cultured with traditional STAT1 stimuli (e.g. IL-27 in Hirahara et al., Immunity 2015 or interferons in Iwata et al., Immunity 2017)doi: 10.1016/j.immuni.2015.04.014, 10.1016/j.immuni.2017.05.005).*

Thank you for your insightful comments. We appreciate your attention to the choice of the STAT1 gene set in our RNA-seq analysis.

Initially, we selected the STAT1 gene set from a study involving IL-21 stimulation (GSE63204) because IL-21 is known to activate STAT1, despite STAT3 being its principal signaling pathway. However, we acknowledge that this choice may not have been optimal given the context of our study, which focuses on the role of IL-27 and its impact on STAT1 signaling in T cells.

We agree that gene sets derived from studies using more canonical STAT1 stimuli, such as IL-27 or interferons, would be more relevant for our analysis. In response to your suggestion, we have revised our approach and adopted a gene set from GSE65621, which compares STAT1-/- and wild-type CD4 T cells following IL-27 stimulation. This gene set is more aligned with the focus of our study and provides a more appropriate reference for identifying STAT1-activated genes.

Our re-analysis revealed that 270 genes (FPKM > 1, log2FC > 2) were downregulated in STAT1-/- cells compared to wild-type cells, which we defined as STAT1-activated genes. Notably, approximately 50% of the upregulated differentially expressed genes (55 out of 137) in our dataset fell into the category of STAT1-activated genes, while none were classified as STAT1-suppressed genes (Figure 4B). Furthermore, Gene Set Enrichment Analysis (GSEA) demonstrated significant enrichment of STAT1-activated genes in the transcriptome of CD4 SP thymocytes from the knockout mice (NES = 1.67, nominal p-value = 10^{-16}, Figure 4D).

These findings support our conclusion that IL-27p28 deficiency leads to enhanced STAT1 activity in CD4 SP thymocytes. We believe that using a more relevant gene set has strengthened our analysis and provided clearer insights into the molecular mechanisms underlying the observed phenotype.

We have cited the relevant studies (Hirahara et al., Immunity 2015; Iwata et al., Immunity 2017) to provide context for our revised analysis and to acknowledge the importance of canonical STAT1 stimuli in T cell signaling. We appreciate your guidance and are confident that these revisions have improved the robustness and relevance of our findings.

> *(5) Given the ability of IL-27 to activate STAT1 in T cells, it is surprising that SP, RTE, and Tn cells from Cd11c-p28-flox mice exhibit more STAT1 signaling than WT controls. If not IL-27, then what is the stimulus for this STAT1 activity? The authors rule out autocrine IFN-g production in vitro (not in vivo) but provide no further insight.*

Thank you for raising this important question. We appreciate your interest in understanding the source of enhanced STAT1 signaling in SP, RTE, and Tn cells from Cd11c-p28-flox mice, especially given the role of IL-27 in activating STAT1 in T cells. As previously discussed, we have identified IL-35 and CLCF1 as the most likely candidate cytokines driving the observed STAT1 activity in the absence of p28. These cytokines are of particular interest due to their potential to activate STAT1 and their relevance in the context of our study.

To address the question of what drives the enhanced STAT1 signaling, we are planning to perform ChIP experiments using these cytokines directly. This approach will allow us to evaluate their impact on STAT1 signaling more precisely, without relying on TCR + co-stimulation. By treating cells ex vivo with IL-35 and CLCF1, we aim to determine whether these cytokines are responsible for the increased STAT1 activity observed in Cd11c-p28-flox mice.

We acknowledge that ruling out autocrine IFN-γ production in vitro, as we have done, does not fully address the potential role of IFN-γ in vivo. Therefore, we are also considering additional in vivo experiments to further investigate this possibility. These studies will help us determine whether other sources of IFN-γ or other cytokines contribute to the observed STAT1 hyperactivation. Unfortunately, due to unforeseen challenges with mouse crosses, we anticipate a delay of approximately six months to obtain the appropriate genotypes necessary for these experiments. We are actively working to resolve these challenges and will update the manuscript with the results of these experiments upon completion.

> *(6) The RNAseq data affirms that SP, RTE, and Tn cells from Cd11c-p28-flox mice exhibit more STAT1 signaling than WT controls. However, this does little to explain the attendant hyper-Th1 phenotype. Is there evidence that epigenetic machinery is deregulated (to account for changes in DNA. histone methylation)? Were IFN-g and Tbet among these few observed DEG? If so, then this should be highlighted. If not, then the authors must address why not. Are there clues as to why STAT1 signing is exaggerated? Also, the hyper-STAT1 effect should be better described using more rigorous STAT1- and interferon-signature genesets (see the work of Virginia Pascual, Anne O'Garra).*

Thank you for your valuable feedback and suggestions. We appreciate your interest in understanding the mechanisms underlying the hyper-Th1 phenotype observed in Cd11c-p28-flox mice. Below, we address each of your points in detail:

(1) Epigenetic Regulation:

We have conducted a thorough analysis of the global levels of key histone modifications, including H3K4me3, H3K9me3, and H3K27me3, as well as the mRNA expression of the enzymes responsible for catalyzing these marks. Our results indicate that there are no significant differences in these histone modifications or the expression of the associated enzymes between Cd11c-*p28*$^{f/f}$ and wildtype mice (Figure 3-figure supplement 1A-C). This suggests that the enhanced STAT1 signaling is not a consequence of broad epigenetic deregulation. Instead, we hypothesize that the observed changes may be driven by more specific molecular mechanisms, such as cytokine signaling pathways.

(2) IFN-γ and Tbx21 Expression:

Regarding the expression of Th1-associated genes, our analysis revealed a modest induction of *ifng* and *tbx21* (encoding T-bet) in the CD4SP population following TCR stimulation. However, the baseline expression levels of these genes were quite low in freshly isolated CD4SP cells. Specifically, *ifng* was undetectable, and *tbx21* had an FPKM of 0.29 in wildtype mice compared to 1.05 in *Cd11c-p28*$^{f/f}$ mice. While these findings indicate some upregulation of Th1-associated genes, the overall expression levels remain relatively low, suggesting that additional factors may contribute to the hyper-Th1 phenotype.

(3) STAT1 Signature Genesets:

We have revised our analysis to incorporate more rigorous STAT1 and interferon-signature genesets, as suggested. We have adopted gene sets from well-established studies, including those by Virginia Pascual and Anne O'Garra, to provide a more comprehensive and accurate assessment of STAT1 signaling. This approach has enhanced our ability to identify and characterize the genes involved in the STAT1 pathway, providing clearer insights into the exaggerated STAT1 signaling observed in our model.

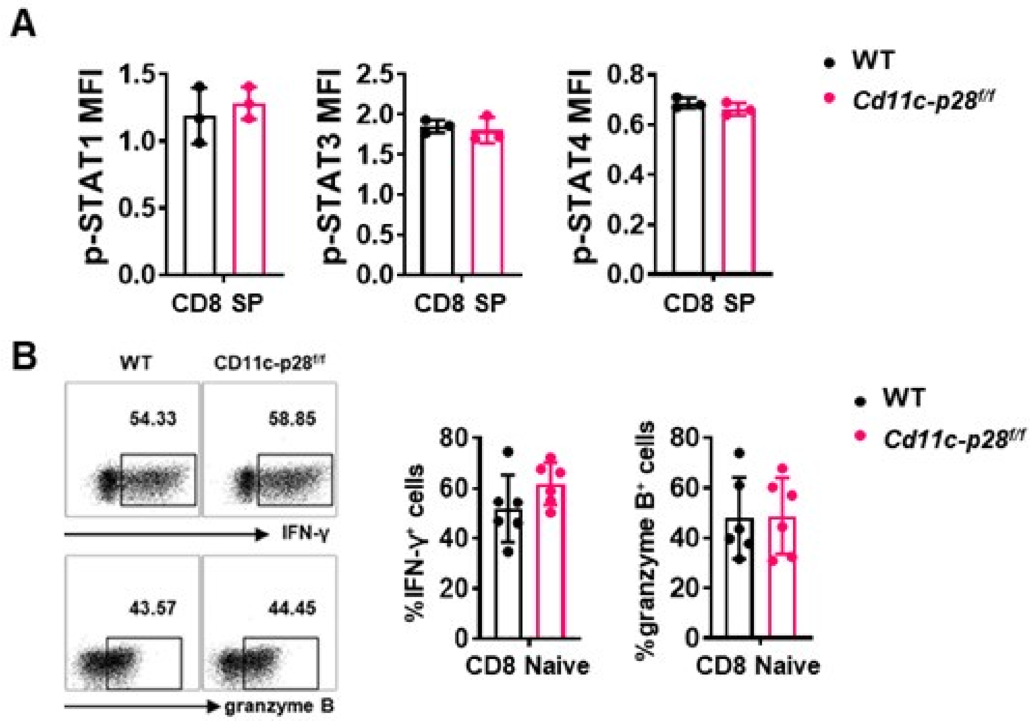
We appreciate your guidance and are committed to refining our analysis to provide a more detailed understanding of the mechanisms driving the hyper-Th1 phenotype in Cd11c-p28-flox mice. We will continue to explore the potential roles of cytokines such as IL-35 and CLCF1, as well as other factors that may contribute to the observed changes in STAT1 signaling and Th1 differentiation. We look forward to sharing our updated findings and further discussing these mechanisms in our revised manuscript.

> *(7) Is the hyper-Th1 phenotype of SP, RTE, and Tn cells from Cd11c-p28-flox mice unique to the CD4 compartment? Are developing CD8$^+$ cells similarly prone to increased STAT1 signaling and IFN-g production?*

Thank you for raising this important point. Our data indeed suggests that the hyper-Th1 phenotype observed in SP, RTE, and Tn cells from *Cd11c-p28*$^{f/f}$ mice is unique to the CD4$^+$ T cell compartment. Specifically, we found that while CD4$^+$ SP cells from *Cd11c-p28*$^{f/f}$ mice exhibited a significant upregulation in IL-27 receptor expression (both IL27Ra and gp130) compared to wild-type (WT) mice, CD8$^+$ SP cells from the same genotype showed markedly lower expression of these receptor subunits (Figure 1C in Sci Rep. 2016 Jul 29:6:30448. DOI: 10.1038/srep30448). This finding is further supported by our observation that the phosphorylation levels of STAT1, STAT3, and STAT4, downstream targets of IL-27 signaling, were comparable between CD8 SP cells from *Cd11c-p28*$^{f/f}$ and WT mice (Author response image 1). Additionally, we observed no significant difference in IFN-γ and granzyme B production between naïve CD8 T cells isolated from the lymph nodes of the two genotypes (Author response image 1). Taken together, these results suggest that the enhanced Th1 differentiation and IFN-γ production seen in the CD4$^+$ T cell population from *Cd11c-p28*$^{f/f}$ mice is not recapitulated in the CD8$^+$ T cell lineage.

Author response image 2.

(A) Intracellular staining was performed with freshly isolated thymocytes from *Cd11c-p28*$^{f/f}$ mice and WT littermates mice using antibodies against phosphorylated STAT1 (Y701), STAT3 (Y705), and STAT4 (Y693). The mean fluorescence intensity (MFI) for CD8 SP from three independent experiments (mean ± SD, n=3). (B) CD8$^+$ naive T cells were cultured under Th0 conditions for 3 days. The frequency of IFN-γ-, and granzyme B-producing CD8$^+$ T cells were determined analyzed by intracellular staining. Representative dot plots (left) and quantification (right, mean ± SD, n=6).



> *Minor points and questions*
>
> *(1) Line 84 - Villarino et al. and Pflanz et al. are mis-referenced. Neither involves Trypanosome studies. The former is on Toxoplasma infection and, thus, should be properly referenced in the following sentence.*

Thank you for pointing out this error. You are correct that the references to Villarino et al. and Pflanz et al. were misapplied in the context of Trypanosome studies. Villarino et al. focuses on Toxoplasma infection, and we appreciate your guidance to ensure accurate citation. We will correct this in the manuscript and properly cite the studies in their appropriate contexts. Thank you for your vigilance in maintaining the accuracy of our references.

> *(2) T—bet protein should also be measured by cytometry*

We sincerely thank the reviewer for the valuable suggestion regarding the measurement of T-bet protein levels. In response to this comment, we have performed additional experiments to quantify T-bet protein expression using flow cytometry. The results of these analyses have been incorporated into the revised manuscript as Figure 1F.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *(1) When new mouse strains are generated in this study, there is no comment on whether there are any changes in the frequency or cell number of CD4 T cells. For instance, in Aire-deficient CD11c-p28 floxed mice, it should be noted whether CD4SP, naïve CD4, and CD4 RTE are all the same in frequency and number compared to their littermate controls. Also, is there any effect on the generation of these thymocytes?*

We sincerely thank the reviewer for raising this important point regarding the potential changes in the frequency and cell numbers of $CD4^+$ T cells in the newly generated mouse strains. In response to the reviewer's question, we would like to clarify the following:

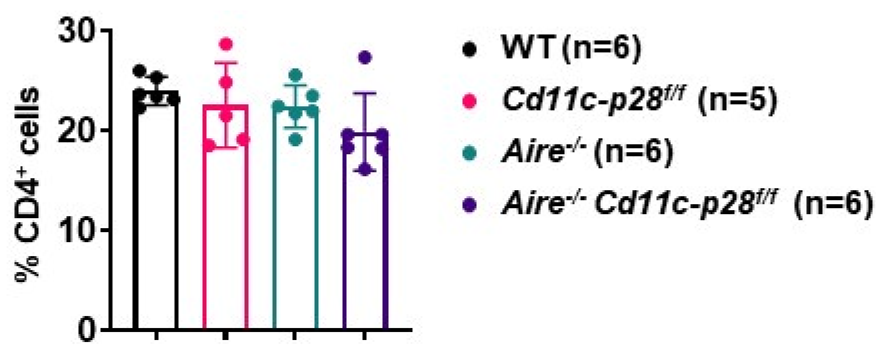
(1) Impact of *Aire* deficiency on $CD4^+$ T Cells:

As previously reported by us and others (Aging Dis. 2019, doi: 10.14336/AD.2018.0608; Science. 2002, doi: 10.1126/science.1075958), *Aire* deficiency does not significantly alter the overall number or frequency of CD4 single-positive (CD4SP) thymocytes, recent thymic emigrants (RTEs), or naïve $CD4^+$ T cells. However, it profoundly affects their composition and functional properties, leading to the escape of autoreactive T cells and subsequent autoimmune manifestations.

(2) Observations in *Cd11c-p28*$^{f/f}$*Aire*$^{-/-}$ mice:

In our study, we observed that the number and frequency of $CD4^+$ T cells in the spleen and lymph nodes were comparable among *Cd11c-p28*$^{f/f}$, *Aire*$^{-/-}$, and *Cd11c-p28*$^{f/f}$*Aire*$^{-/-}$ mice, and WT controls. This suggests that the genetic modifications did not significantly impact the overall development or peripheral maintenance of $CD4^+$ T cells.

Author response image 3.



(3) Challenges in assessing RTEs in double knockout mice:

To accurately assess RTEs in the double knockout mice, it would be necessary to cross these mice with Rag-GFP reporter mice, which specifically label RTEs. However, breeding the appropriate mouse strain for this analysis would require additional time and resources, which were beyond the scope of the current study.

> *(2) There are a couple of typos throughout the manuscript. For example, line 91: IL-27Rα or line 313: phenotype.*

We apologize for the typographical errors. We have carefully reviewed the entire manuscript and corrected all identified mistakes, including those on line 91 (IL-27Rα) and line 305 (phenotype).

> *(4) The authors should show each data point on their bar graphs.*

Thank you for the suggestion. We have presented each data point on their bar graphs in the revised manuscript.

> *(4) It should be noted from which organs the RTE and the naïve T cells were harvested.*

Thank you for the constructive suggestion. We isolated $CD4^+$ RTEs and mature naive $CD4^+$ T cells by sorting $GFP^+CD4^+CD8^-CD^-NK1.1^-$ cells (RTEs) and $GFP^-CD4^+CD8^-CD^-CD44^{lo}$ cells (naive T cells) from lymph nodes. This detail has been added to the manuscript on line 475.

https://doi.org/10.7554/eLife.96868.2.sa3